

MATH40000 Double Project
Evaluating Significance Tests in Generalised
Additive Models

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Abstract

Generalised additive models extend the generalised linear model framework by allowing covariate effects to be represented by smooth functions, giving greater flexibility in modelling nonlinear relationships. In the ‘mgcv’ package in R, approximate significance tests and associated p -values are reported for these smooth terms, but they rely on several asymptotic assumptions and heuristic approximations. This project reviews the framework behind generalised additive models, explains how these tests are constructed, and evaluates their reliability through empirical simulation studies of type I error calibration and power across a range of modelling scenarios. Particular attention is given to the effects of response family, sample size, smoothing parameter estimation method, and model complexity in order to extend existing empirical assessments of smooth term inference from Wood (2013) and Young et al. (2011). The results indicate that the tests are generally well behaved in relatively simple settings, particularly for moderate sample sizes and when REML smoothing parameter estimation method is used, with some inflation at higher nominal levels. However, noticeable inflation in the p -values occurs when the model must distinguish between linear and nonlinear effects of the same covariate. An additional permutation comparison was implemented in this most challenging Gaussian setting, which substantially reduced this inflation, suggesting that the approximate reference distribution used by ‘mgcv’ contributes meaningfully to the miscalibration seen in this setting.

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1 Introduction

Generalised additive models (GAMs) provide a flexible extension to the generalised linear model (GLM) framework by allowing the effects of covariates on the response to be represented by smooth functions rather than restricted to parametric terms. This makes GAMs particularly useful where relationships between variables are expected to be nonlinear, whilst retaining much of the familiar inference and structure from GLMs through link functions and exponential family response distributions.

This flexibility has made GAMs an increasingly popular choice in applied modelling. For example, Ravindra et al. (2019) describe their effective use in environmental epidemiology to investigate associations between air pollution exposure and climate change with human health. The smooth functions allow for much more complex relationships between variables to be identified when compared to the parametric approach in GLMs. Furthermore, Doohan et al. (2026) found across 430 real world datasets that GAMs remain competitive against neural networks for prediction tasks whilst maintaining greater interpretability, especially in smaller data sample settings. In such applications, smooth terms are often interpreted directly as evidence of nonlinear relationships, making the reliability of their associated significance tests an important practical consideration.

In practice, statistical software can fit GAMs to data and provide inferential summaries for the smooth terms included in the model. In particular, the ‘mgcv’ package in R, which was developed by Simon N. Wood (2001), reports p -values associated with the approximate significance tests for each smooth component fitted to the model, allowing users to quickly assess whether a variable gives a meaningful contribution to the response. We focus here on the approximate tests implemented through the ‘mgcv’ package due to its popularity and well established implementation. However, more so than GLMs, these significance tests rely on several layers of approximation, including approximate normality of estimators, Bayesian covariate arguments, effective degrees of freedom and the treatment of smoothing parameters as fixed when deriving the distributional results for the significance tests. The theoretical and computational basis underlying the ‘mgcv’ package is described by Wood (2006a), while details of the R package implementation are available on the CRAN package page, by Wood (2025).

This raises an important inferential question: how reliable are the reported p -values for smooth terms in GAMs? Since these values are used in practical interpretation of the smooth terms, it is important to understand the extent to which they provide trustworthy inference, and in particular whether the behaviour deteriorates in modelling settings where the underlying approximations may be less accurate.

The approximate significance tests reported for smooth terms in GAMs have been examined and critically evaluated through theoretical and empirical study. In particular, Wood (2013) shows that the p -values in ‘mgcv’ can provide effective zero effect and nonzero effect tests in GAMs for a range of relatively simple settings. Wood’s focus is primarily on the methodology and justification behind the construction of the p -values, rather than exploring a wide range of modelling complications, such as different response families or smoothing parameter estimation methods. Empirical concerns in the p -values are raised by Young et al. (2011), who demonstrate that significance tests in LOESS-based GAMs exhibit inflated type I error, particularly when smoothing parameters are selected from the observed data. They found that unconditional permutation tests may provide more effective significance test calibration. Their findings are relevant to our study since they show that data-driven smoothing selection can distort nominal significance levels in additive models. However, their setting differs from the penalised GAM framework used in ‘mgcv’. LOESS smooth terms are locally weighted regression estimators, a generalisation of the least-squares method as stated by Jacoby (2000), whereas ‘mgcv’ represents smooth functions using basis expansions and controls complexity using explicit smoothing penalties. Therefore, Young et al. provide important motivation for studying smooth term inference, but their results cannot be assumed to transfer directly to the penalised tests considered here. Furthermore, the work by Young et al. primarily focuses on type I error inflation in the LOESS-based setting in binomial and Gaussian settings. This motivates a broader empirical study of the behaviour of the ‘mgcv’ tests under controlled simulation scenarios.

Motivated by this gap, the aim of this project is to provide a review of the general framework of penalised GAMs, explain how significance tests for smooth terms are constructed, and evaluate their behaviour empirically under controlled conditions. The simulation study is designed specifically to address the approximations found in the construction of the significance

tests and to address the limitations of the existing literature discussed above. To extend Wood’s relatively simple empirical settings, the simulations vary response distribution, smoothing parameter estimation method, sample size and model complexity. To extend Young et al.’s focus on type I error in LOESS-based GAMs, the simulations instead consider penalised GAMs in ‘mgcv’, consider both type I error and power, and include settings with multiple smooths and correlated covariates. We also include a residual permutation comparison in the most challenging Gaussian setting to assess whether an empirical null distribution can improve calibration relative to the approximate ‘mgcv’ reference distribution.

Overall, the results suggest that the approximate significance tests are reasonably well calibrated in relatively simple settings with moderate sample size, although some inflation remains at nominal levels towards the higher end of the range $p \leq 0.10$. The simulations indicate that REML generally provides more reliable inference than GCV in the simple scenarios considered, and that the most substantial difficulties arise in settings where the model must distinguish between a true linear effect and an additional smooth contribution for the same covariate. In these cases, type I error can remain noticeably inflated, and power must be interpreted relative to this already inflated null rejection rate. In the Gaussian case, the residual permutation test explored relative to this stressed scenario substantially reduces inflation, suggesting that the approximate reference distribution used by ‘mgcv’ is an important source of miscalibration in this setting. This motivates further study of alternative inferential procedures, particularly for non-Gaussian responses and for tests of nonlinearity beyond an already included linear effect.

The next chapter develops the general framework of generalised additive models required for the later study of significance testing. In particular, it introduces spline basis representation of smooth functions, the role of penalised likelihood in model fitting, the selection of smoothing parameters and the effective degrees of freedom used to describe model complexity. This provides understanding required to explain how approximate significance tests for smooth terms are constructed in ‘mgcv’.

2 Generalised Additive Models Framework

This chapter introduces the framework of generalised additive models needed to understand the construction of the significance tests for smooth terms. Much of this framework is outlined by Wood (2006a). We assume familiarity with the framework of generalised linear models, as outlined by Dobson and Barnett (2018).

2.1 Extending from GLMs to GAMs

Suppose we have a dataset $(\mathbf{x}_i, y_i), i = 1, \dots, n$ with n observations, where y_i is the response variable and $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^T$ is the vector of p predictor variables for the i th observation. Our aim is to model the relationship between the response variable and the predictor variables.

In the generalised linear model framework, this relationship is expressed as

$$g(\mu_i) = \mathbf{x}_i \boldsymbol{\beta},$$

where $\mu_i = \mathbb{E}[y_i | \mathbf{x}_i]$, $\boldsymbol{\beta}$ is the vector of parameters to be estimated and g is the specified link function. The link function maps the conditional mean of the response, whose range is determined by the chosen distribution, onto the real valued scale of the linear predictor.

A generalised additive model extends this framework by replacing linear covariate effects with smooth functions. This model is then written as

$$g(\mu_i) = c + \sum_{j=1}^p f_j(x_{ij}),$$

where f_j are unknown smooth functions of the covariates and c is an intercept term. This allows the relationship between the predictors and the response to be modelled more flexibly whilst retaining the general structure of a GLM, which assists in parameter estimation and inference.

In order to fit a GAM in practice, smooth functions f_j must be represented in a form that can be estimated from the data. We therefore begin by considering how these smooth functions can be constructed.

2.2 Spline regression

Among the various nonparametric smoothing methods available, we focus on spline regression because of its interpretability and its implementation in the ‘mgcv’ package. For simplicity, we will focus on univariate smoothing here, although the same concepts can be extended to multiple covariates in GAMs.

To estimate a smooth function f in a GAM, we can represent it using a basis expansion. Given b_j denotes the j th basis function, we can express f as

$$f(x) = \sum_{j=1}^k \beta_j b_j(x),$$

where k is the basis dimension, that is, the number of basis functions used to estimate f , and β_j are the coefficients to be estimated from the data (Wood; 2006a, pp. 146–161).

In practice, spline bases are commonly used as they offer a flexible and computationally efficient way to represent smooth functions. A spline is a piecewise polynomial, where each sub-polynomial is defined on intervals $[t_i, t_{i+1})$ with the endpoints t_i called knots. These sub-polynomials are joined such that the polynomial remains smooth at the knots. The order of the spline is defined by the highest degree of the polynomial used within the piecewise polynomial.

Using splines to estimate smooth functions, it has been proven (Härdle et al.; 2004, pp. 101–103) that the smoothest interpolant for a given set of data points is the cubic spline. To show this in a similar fashion to Härdle et al., we first consider observations (x_i, y_i) where $i = 1, \dots, n$, and suppose we wish to estimate a smooth function f relating x to y . A natural starting point is the residual sum of squares, given by

$$\text{RSS}(f) = \sum_{i=1}^n \{y_i - f(x_i)\}^2.$$

If we minimised this quantity alone, we would allow the fitted function to follow the data as closely as possible. However, in the nonparametric approach, this can lead to an overfitting that may have poor out-of-sample results. To control this, we add a stabiliser to the residual sum of squares,

which penalises the roughness of $f(x)$. A common stabiliser is given by

$$\|f''\|_2^2 = \int \{f''(x)\}^2 dx,$$

which is particularly effective as it directly measures the curvature of the function, a key indicator of its roughness.

In order to measure how much we want to penalise the function for its roughness, we introduce the smoothing parameter λ . This directly controls the trade-off between overfitting and underfitting the data. Redefining the residual sum of squares with the stabiliser, we have

$$S_\lambda(f) = \sum_{i=1}^n \{y_i - f(x_i)\}^2 + \lambda \|f''\|_2^2.$$

It is clear that the larger λ is, the smoother the spline estimate; as $\lambda \rightarrow 0$, the minimiser perfectly interpolates the response y , whilst as $\lambda \rightarrow \infty$, the penalty dominates and the minimiser approaches a straight line.

Continuing with the minimisation problem, the minimiser of f , denoted

$$\hat{f}_\lambda = \arg \min_f S_\lambda(f),$$

is the spline regressor. It was shown by Härdle et al. that when the minimisation is taken over all twice continuously differentiable functions, the unique minimiser for fixed λ is a natural cubic spline estimator with knots at the ordered observed values of x . Hence, the solution can be represented piecewise by cubic polynomials of the form

$$p_i(x) = \alpha_i + \beta_i x + \gamma_i x^2 + \delta_i x^3,$$

where $i = 1, \dots, n - 1$ and each p_i is used independently between adjacent knots. This result provides the theoretical motivation for representing smooth functions using spline bases.

In the ‘mgcv’ package, cubic spline regression can be used to estimate each smooth function in a GAM. However, this is not its default method. Instead, it uses a thin-plate regression (Wood; 2006a, pp. 154–160) as it addresses some limitations of cubic spline regression, such as the choice of knot placement and which choice of basis may be better suited to the data.

The basis dimension k may be specified by the user or chosen by default by R, and so it remains to consider how to choose an appropriate value for k in order to balance the flexibility of the spline model and computational efficiency.

2.3 Determining the basis dimension

As mentioned previously, the basis dimension k is the number of basis functions used to estimate the smooth function f . Generally, a larger basis dimension allows the spline basis to capture more flexibility and complexity in the data, but at an increased computational cost. Choosing an appropriate value for k must ensure that the basis is flexible enough to sufficiently capture the underlying function without overfitting the data.

In practice, the exact size of the basis dimension is not a value of significant importance, as proposed by Wood (2006a, p. 161). Rather, k should be chosen as an upper bound of the desired flexibility of the model. If k is sufficiently large, then the fitted model is not typically constrained by the basis dimension itself. Instead, the complexity of the smooth is controlled mainly by the effective degrees of freedom (EDF) and the smoothing parameter λ . This is because some basis coefficients may be shrunk towards zero, so that the fitted smooth effectively uses fewer degrees of freedom than the basis dimension would suggest. Both of these concepts will be further explored later in this chapter.

However, there do exist some cases in which the model is sensitive to basis dimension, such as if the basis dimension is too small to sufficiently capture the underlying structure of the function. In these cases, we can informally check the model fit for multiple models with varying basis dimensions and see how the model deviance varies across fits.

2.4 Matrix representation of a GAM

We now move from the construction of individual smooth functions to the form of the full generalised additive model, derived in a style similar to Wood (2006a, pp. 167–168). We begin by recalling the general form of the GAM:

$$g(\mu_i) = c + \sum_{j=1}^p f_j(x_{ij}), \quad i = 1, \dots, n$$

where each smooth function is assumed to be constructed from a single covariate, p is the number of smooth functions, $\mu_i = \mathbb{E}[y_i|\mathbf{x}_i]$, and g is a known monotonic and twice differentiable link function. We also assume that response y_i follows a distribution from the exponential family.

For simplicity, we first consider the case $p = 1$, $c = 0$ and g is the identity link, such that

$$\mu_i = f(x_i).$$

As seen previously, each smooth function is represented using a basis expansion of the form

$$f(x_i) = \sum_{j=1}^k \beta_j b_j(x_i)$$

where coefficients β_j are estimated in the model fitting.

Defining the $n \times k$ basis matrix $\mathbf{B}$ with entries $\mathbf{B}_{cd} = b_d(x_c)$, the $k \times 1$ vector of coefficients

$$\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_k)^\top,$$

and the $n \times 1$ mean vector

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^\top,$$

we can redefine this model in matrix form as

$$\boldsymbol{\mu} = \mathbf{B}\boldsymbol{\beta}.$$

We now extend this to the full GAM, whilst still assuming identity link function. First, we redefine the notation for the basis functions in the construction of our smooth functions, such that the j th smooth can be written as

$$f_j(x_{ij}) = \sum_{a=1}^{k_j} \beta_{ja} b_{ja}(x_{ij}).$$

Applying the basis matrix representation to each smooth term used in the

model gives

$$\boldsymbol{\mu} = c\mathbf{1}_n + \sum_{j=1}^p \mathbf{B}_j \boldsymbol{\beta}_j,$$

where $\mathbf{B}_j \boldsymbol{\beta}_j$ represents the contribution of the j th smooth function. This can be more compactly written by defining the model matrix

$$\mathbf{X} = (\mathbf{1}_n, \mathbf{B}_1, \dots, \mathbf{B}_p)$$

of dimension $n \times (1 + \sum_{j=1}^p k_j)$, and the parameter matrix

$$\boldsymbol{\beta} = (c, \boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_p^\top)^\top$$

of dimension $(1 + \sum_{j=1}^p k_j) \times 1$. Hence, the GAM takes the simpler matrix form

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}.$$

This representation is similar to that of a GLM. However, unlike the GLM case, we cannot estimate the vector of parameters based on ordinary least squares, since each smooth term is subject to penalisation.

2.5 Penalised likelihood

In a GLM, parameter estimates are obtained by maximising the likelihood function, $l(\boldsymbol{\beta})$. In a GAM, estimation must additionally control the smoothness of nonparametric functions to avoid overfitting. This is achieved through penalised likelihood. Rather than maximising the likelihood alone, we maximise a modified criterion that balances goodness of fit with smoothness of the estimated functions.

To express this penalty in matrix form, assuming cubic spline regression, we consider the roughness measure for the j th smooth function,

$$\|f_j''\|_2^2 = \int \{f_j''(x)\}^2 dx.$$

We redefine this using the basis representation of the smooth.

$$\int \{f_j''(x)\}^2 dx = \int \left\{ \sum_{a=1}^{k_j} \beta_{ja} b_{ja}''(x) \right\}^2,$$

where we apply the twice differentiation with respect to x to each basis function since f_j is a linear combination of functions of x . We can redefine this in matrix form as

$$\int \{f_j''(x)\}^2 dx = \boldsymbol{\beta}_j^\top \mathbf{S}_j \boldsymbol{\beta}_j,$$

where we define $\boldsymbol{\beta}_j$ as the $k_j \times 1$ vector of coefficients of the basis functions for the j th smooth function and $\mathbf{S}_j$ as the corresponding $k_j \times k_j$ penalty matrix with entries

$$[\mathbf{S}_j]_{ab} = \int b_{ja}''(x) b_{jb}''(x).$$

Using a smoothing parameter λ_j to determine the roughness of the j th function and finding the total penalisation term over all smooth functions used in the model, Wood (2006a, pp. 167–169) derives the penalised likelihood expressed as

$$l_p(\boldsymbol{\beta}) = l(\boldsymbol{\beta}) - \frac{1}{2} \sum_{j=1}^p \lambda_j \boldsymbol{\beta}_j^\top \mathbf{S}_j \boldsymbol{\beta}_j.$$

Therefore, larger values of λ_j impose a stronger penalty on roughness for the smooth, producing smoother fitted functions. To write this more compactly, we reintroduce the full parameter vector $\boldsymbol{\beta}$ defined in the previous section. If we redefine each $\mathbf{S}_j$ to be embedded into a larger matrix $\bar{\mathbf{S}}_j$ by padding with zeros such that

$$\boldsymbol{\beta}^\top \bar{\mathbf{S}}_j \boldsymbol{\beta} \equiv \boldsymbol{\beta}_j^\top \mathbf{S}_j \boldsymbol{\beta}_j,$$

then, defining

$$\mathbf{S} = \sum_{j=1}^p \lambda_j \bar{\mathbf{S}}_j,$$

the penalised likelihood becomes

$$l_p(\boldsymbol{\beta}) = l(\boldsymbol{\beta}) - \frac{1}{2} \boldsymbol{\beta}^\top \mathbf{S} \boldsymbol{\beta}.$$

This is the form used in GAM model fitting in ‘mgcv’. The presence of penalty matrix $\mathbf{S}$ means that estimation no longer depends only on the likelihood, but also on the smoothing parameters and the structure of the penalty matrix. This is one of the key reasons why inference for smooth terms is far more complicated than for ordinary parametric terms in a GLM.

Estimation of β can then be carried out by penalised iteratively re-weighted least squares.

2.6 Fitting a GAM via P-IRLS

For fixed smoothing parameters, the process of fitting a GAM in ‘mgcv’ can be carried out using penalised iteratively reweighted least squares (P-IRLS), derived from the penalised likelihood, as outlined by Wood (2001, pp. 21–22).

Suppose the smoothing parameters, and hence the penalty matrix $\mathbf{S}$, are fixed. Given a current estimate of β , denoted $\beta^{[q]}$, we obtain current estimates of μ and V_i , the variance of each y_i . From these quantities, we define the diagonal weight matrix $\mathbf{W}$, with diagonal entries

$$W_{ii} = [g'(\mu_i)^2 V_i]^{-1},$$

and the pseudodata vector $\mathbf{z}$, given by

$$\mathbf{z} = \mathbf{X}\beta + \mathbf{\Gamma}(\mathbf{y} - \mu),$$

where diagonal matrix $\mathbf{\Gamma}$ has entries

$$\mathbf{\Gamma}_{ii} = [g'(\mu_i)]^{-1}.$$

Then, for fixed $\mathbf{S}$, the updated estimate of β is obtained by minimising the penalised weighted least squares criterion,

$$\|\mathbf{W}^{1/2}(\mathbf{z} - \mathbf{X}\beta)\|^2 + \beta^\top \mathbf{S}\beta$$

with respect to β . Whilst the penalty term is quadratic in β , the weighted least square term arises from a local approximation to the likelihood for non-Gaussian responses, which may reduce the accuracy of the resulting approximation.

In practice, the smoothing parameters are not typically fixed and are estimated from the data. There are several methods by which smoothing parameters can be estimated, including generalised cross-validation and restricted maximum likelihood, which will be explored further later in this chapter.

2.7 Effective degrees of freedom

To understand the complexity of a fitted GAM, we introduce the influence matrix

$$\mathbf{A} = \mathbf{X}(\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S})^{-1} \mathbf{X}^\top \mathbf{W},$$

which maps the vector of responses $\mathbf{y}$ to the fitted values of the model. This matrix plays a similar role to the hat matrix for a GLM,

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top,$$

where we state the hat matrix with identity weight. The trace of this matrix is equal to the degrees of freedom of the model. In a GAM, this cannot carry across directly to the matrix $\mathbf{A}$ as the smooth terms are penalised and dependent on the smoothing parameter, and so do not behave as fully free parameters.

We therefore define the effective degrees of freedom (EDF) of the fitted model by

$$\text{EDF} = \text{tr}(\mathbf{A}).$$

This measures the effective complexity of the fitted smooths. For example, if the EDF for a smooth is approximately equal to 1, then the smooth will behave similarly to if it were replaced with a linear term of the same covariate. Larger EDF generally corresponds to greater flexibility in the fitted smooth, although the EDF remains constrained by the chosen basis dimension if it is limiting.

It remains to explain how the smoothing parameters are estimated in the penalty matrix $\mathbf{S}$.

2.8 Selection of smoothing parameter

The smoothing parameters λ_j control the trade-off between bias and variance, by balancing closeness to the observed data against the smoothness of the fitted functions. Their choice is therefore an important part of fitting a GAM. In practice, the optimal smoothing parameters are estimated from the data rather than chosen in advance. This estimated optimum is then used in penalised likelihood and treated as known in further inference.

One approach to estimating the smoothing parameters is by cross-validation. The ordinary cross-validation (OCV) score, as stated by Wood (2006a,

pp. 172–177), is given by

$$CV_O = \frac{1}{n} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{(1 - A_{ii})^2},$$

where $\hat{\mu}_i$ is the fitted mean from the data for the i th observation, and $\mathbf{A}$ is the influence matrix. This criterion measures predictive performance over every single data point, and therefore is likely to be computationally expensive.

To reduce the cost of this method, ‘mgcv’ instead uses the generalised cross-validation (GCV) score,

$$CV_G(\lambda) = \frac{n \|\mathbf{y} - \hat{\boldsymbol{\mu}}\|^2}{[n - \text{tr}(\mathbf{A})]^2}.$$

This replaces the individual weights in the OCV denominator by an average measure of model complexity using the effective degrees of freedom, $\text{tr}(\mathbf{A})$.

GCV is therefore a computationally efficient approximation to ordinary cross-validation. However, the resulting smoothing parameter estimates will also be approximate. This is relevant for later inference, since the significance tests in ‘mgcv’ treat the smoothing parameters as fixed after estimation, so any inaccuracy in their selection may affect the calibration of the reported p -values.

An alternative approach to estimating the smoothing parameters is restricted maximum likelihood (REML). Following Wood (2011), this method treats smooth functions as random effects, so that the smoothing parameters λ_j can be interpreted as variance parameters to be estimated by likelihood-based criterion. This differs from GCV, which instead selects smoothing parameters by minimising model prediction error. Given the different criterion used by each method, it is natural to investigate whether the methods provide different inferential behaviour for the approximate smooth term p -values, which will be explored in simulation studies.

2.9 Implications for significance testing

The framework developed in this chapter shows that, although GAMs can be written in a form similar to GLMs, model fitting and parameter estimation for smooth terms is more complicated. Smooth functions are estimated through penalised likelihood, their flexibility depends on smoothing param-

eters estimated from the data, and model complexity is described through effective rather than exact degrees of freedom.

As a result, the significance tests reported for smooth terms cannot be justified in the same direct way as the usual parametric tests in GLMs. Despite this, the ‘summary’ output in ‘mgcv’ provides p -values for each smooth function. It is therefore necessary to understand how the approximate distributions underlying these tests are constructed and which assumptions are required for them to provide reliable inference. The next chapter therefore considers how these approximate test statistics are constructed in ‘mgcv’ and identifies the main sources of approximation that may affect the reliability of the resulting p -values.

3 Significance Testing in GAMs

Using the established framework behind GAMs, significance testing for smooth components remains challenging. For a GAM of the form

$$g(\mu_i) = c + \sum_{j=1}^p f_j(x_{ij}), \quad i = 1, \dots, n,$$

the hypothesis of interest for the j th smooth is

$$H_0 : f_j(x) \equiv 0,$$

that is, the j th smooth function has no effect on the response on the scale of the linear predictor. The ‘mgcv’ package reports p -values for smooth terms via χ^2 or F -tests, but these rely on several layers of approximation. This includes normality assumptions, the use of a Bayesian covariance matrix, the role of effective degrees of freedom, and the treatment of smoothing parameters as fixed after estimation.

In this chapter, we explain how these approximate test statistics are constructed in ‘mgcv’ and identify the assumptions on which they depend. This provides the theoretical basis for the simulation studies that follow, where we assess the extent to which the reported p -values are well calibrated and the settings in which they may become unreliable.

3.1 Normal approximation for smooth term inference

To construct significance tests for smooth terms, we first need to quantify the uncertainty in the estimated coefficient vector β .

Within the framework we have developed in the previous chapter, $\hat{\beta}$ is obtained by minimising the penalised likelihood, giving

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y},$$

for data vector $\mathbf{y}$, weight matrix $\mathbf{W}$ and penalty matrix $\mathbf{S}$. We can write the covariance matrix of the data $\mathbf{y}$ as $\mathbf{W}^{-1}\phi$, where ϕ is the dispersion parameter which is different for each distribution in the exponential family. Using this, Wood (2006a, pp. 189–194) found that the covariance matrix of

the estimators $\hat{\boldsymbol{\beta}}$ is

$$\mathbf{V}_e = (\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S})^{-1} \phi,$$

where, if unknown, ϕ is estimated by

$$\hat{\phi} = \frac{\|\mathbf{W}^{1/2}(\mathbf{y} - \hat{\boldsymbol{\mu}})\|^2}{\text{tr}(\mathbf{I} - \mathbf{A})}.$$

This gives the approximate distribution

$$\hat{\boldsymbol{\beta}} \sim N\left(\mathbb{E}[\hat{\boldsymbol{\beta}}], \mathbf{V}_e\right).$$

This result is true for Gaussian responses, and is approximately true for other distributions in the exponential family by the central limit theorem. Therefore, we would require a sufficiently large sample size for this approximation to hold in these cases. This result is particularly useful in testing whether the vector of parameters is zero as this means our approximate normal distribution will be centred about $\mathbf{0}$ under the null hypothesis. However, since generally $\mathbb{E}[\hat{\boldsymbol{\beta}}] \neq \boldsymbol{\beta}$, this frequentist approximation would be biased and unreliable for significance testing. Therefore, we look towards a Bayesian approach to the uncertainty estimation, which is directly used in ‘mgcv’.

Wood outlines that the penalty term can be viewed as imposing a prior onto $\boldsymbol{\beta}$, so that $\hat{\boldsymbol{\beta}}$ can be interpreted as the posterior mean of $\boldsymbol{\beta}$ (Wood; 2013, pp. 221–222). Under this framework, and assuming identity link,

$$\boldsymbol{\beta}|\mathbf{y} \sim N(\hat{\boldsymbol{\beta}}, \mathbf{V}_\beta),$$

where the posterior variance of the vector of coefficients is given by

$$\mathbf{V}_\beta = (\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S})^{-1} \phi.$$

For Gaussian responses, this normal result is exact, since both the likelihood of $\boldsymbol{\beta}$ and the penalty term are quadratic in $\boldsymbol{\beta}$. However, for non-Gaussian responses, the likelihood is replaced by a local quadratic approximation through P-IRLS. In this case, we work with the pseudodata

$$\mathbf{z} = \mathbf{X}\boldsymbol{\beta} + \mathbf{G}(\mathbf{y} - \boldsymbol{\mu}), \quad \mathbf{v} = \mathbf{X}^\top \mathbf{W} \mathbf{z},$$

where diagonal matrix $\mathbf{G}$ has entries $G_{ii} = g'(\mu_i)$. This leads to the approximate normal distribution

$$\boldsymbol{\beta}|\mathbf{v} \sim N\left(\left[\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S}\right]^{-1} \mathbf{v}, \mathbf{V}_\beta\right),$$

where $\mathbf{v}$, $\mathbf{W}$ and the smoothing parameters are evaluated at convergence of their respective fitting algorithms (P-IRLS and GCV/REML). An estimate of ϕ should also be substituted if necessary.

The key point is that the normal distributions used by ‘mgcv’ are exact only in the Gaussian case, and otherwise depend on the local quadratic approximation to the likelihood. Furthermore, the distributions for all responses rely on the treatment of smoothing parameters as fixed after estimation. These approximations are carried forward into the approximate smooth term significance tests.

3.2 Approximate χ^2 and F -tests

Using the normal approximation from the previous section, we now consider how the significance tests reported in ‘mgcv’ are constructed for individual smooth terms. For the j th smooth function, the null hypothesis is

$$H_0 : f_j(x) \equiv 0,$$

which is equivalent to testing whether the corresponding subset of coefficients, denoted $\boldsymbol{\beta}_j$, is identically zero. Under this null hypothesis, we take

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_j] \approx \mathbf{0}.$$

Let $\mathbf{V}_j$ denote the covariance matrix of the estimated coefficient vector $\hat{\boldsymbol{\beta}}_j$ such that $\mathbf{V}_j$ is extracted directly from $\mathbf{V}_e$. For Gaussian responses with identity link, and approximately for other response distributions and link functions, we have

$$\hat{\boldsymbol{\beta}}_j \sim N(\mathbf{0}, \mathbf{V}_j).$$

Wood (2006a, pp. 194–195) states that if $\mathbf{V}_j$ is of full rank, then standard normal distribution results imply that under the null hypothesis,

$$\hat{\boldsymbol{\beta}}_j^\top \mathbf{V}_j^{-1} \hat{\boldsymbol{\beta}}_j \sim \chi_{k_j}^2,$$

where k_j is the number of basis functions used in the construction of the smooth function f_j .

However, in practice, $\mathbf{V}_j$ is often not of full rank since smooth terms are typically heavily penalised. In this case, ‘mgcv’ instead constructs the rank k_j pseudoinverse of the covariance matrix $\mathbf{V}_j$, denoted $\mathbf{V}_j^-$, such that under the null hypothesis, the test statistic becomes

$$\hat{\boldsymbol{\beta}}_j^\top \mathbf{V}_j^- \hat{\boldsymbol{\beta}}_j \sim \chi_{k_j}^2.$$

If an unknown scale parameter ϕ must also be estimated, then a further approximation is required, which leads to the corresponding approximate F -test statistic.

$$\frac{\hat{\boldsymbol{\beta}}_j^\top \mathbf{V}_j^- \hat{\boldsymbol{\beta}}_j / k_j}{\hat{\phi} / \phi} \sim F_{k_j, n-\text{edf}}.$$

This final approximate F -distribution is used to construct the reported p -values for smooth terms in ‘mgcv’. The validity of this reference distribution therefore relies on the normal approximation for $\hat{\boldsymbol{\beta}}_j$, the estimated covariance matrix, and the treatment of smoothing parameters and effective degrees of freedom during model fitting.

3.3 Sources of approximation in p -values

The previous section provides a practical reference distribution for constructing smooth term significance tests. However, this distribution relies on several approximations which must be kept in mind when interpreting the resulting p -values.

First, the tests rely on approximate normality of the estimated coefficient vector. For Gaussian responses, this is exact, but for non-Gaussian responses, it depends on large sample arguments and on the local quadratic approximation used within the P-IRLS algorithm. Second, the covariance matrix used to construct the test statistics treats the smoothing parameters as fixed after estimation, so the additional uncertainty arising from smoothing parameter selection is ignored. Third, quantities such as effective degrees of freedom and, where necessary, the scale parameter are themselves estimated from the data and then inserted into the reference distribution.

Taken together, these approximations suggest that the reported p -values may not be equally reliable in all modelling settings. In particular, their

calibration may deteriorate in smaller samples, for non-Gaussian responses, or when the estimation of smoothing parameters is itself unstable or sensitive to the fitting criterion. These are precisely the issues explored in the simulation studies that follow.

The next chapter therefore investigates these issues empirically through simulation studies of type I error calibration, with the aim of identifying the settings in practice where the approximate significance tests remain reliable and those in which their calibration begins to deteriorate.

4 Type I Error Calibration Study

The previous chapter showed that the smooth term significance tests reported by ‘mgcv’ rely on several approximations, including approximate normality of the estimated coefficients, the treatment of smoothing parameters as fixed after estimation, and the use of estimated effective degrees of freedom in reference distributions. The aim of this chapter is to assess, through simulation, how well these approximate tests control type I error across a range of modelling settings. In particular, we investigate whether the reported smooth term p -values remain well calibrated when key features of the fitted GAM are varied, including response family, sample size, smoothing parameter estimation method, basis dimension, and dependence structure between covariates.

4.0.1 Simulation framework

To evaluate type I error calibration, the simulation framework follows the general approach used by Wood (2013), in which repeated datasets are generated under different data generating processes (DGPs) and the empirical behaviour of the resulting approximate smooth term p -values from the fitted GAM is assessed across simulation replications. Specifically in this study, we generate m independent datasets, each of sample size n , extract the p -value from the model summary of the GAM function in ‘mgcv’ and examine their empirical performance.

Suppose $p_1, \dots, p_m$ denote the p -values obtained from the m fitted models. For a nominal significance level α , the empirical rejection frequency is defined as

$$\hat{F}_\alpha = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[p_j \leq \alpha],$$

where $\mathbf{1}$ is the indicator function. If the significance test is correctly calibrated under the null hypothesis, which in this case is that the smooth function does not truly contribute to the linear predictor, the p -values should follow a $\text{Uniform}(0, 1)$ distribution. Hence, the empirical rejection rate $\hat{F}_\alpha$ should be close to the nominal level α . Systematic deviation of $\hat{F}_\alpha$ above the nominal level indicated liberal inference and inflated type I error, whereas deviation below the nominal level indicates conservative inference.

In addition to these rejection rates, the mean and standard deviation of

the effective degrees of freedom of each smooth term are also reported across simulation replications, which provides a measure of the complexity of the estimated smooth function and gives additional insight into the behaviour of the fitted GAM beyond p -values alone. The EDF also directly impacts the constructed significance tests we are assessing due to their inclusion in the constructed F -tests, providing further insight into which approximations cause deterioration. For some scenarios, empirical cumulative distribution function plots of the simulated p -values is provided over the range $p \leq 0.10$ since this lower tail is the most relevant region for assessing a test's behaviour at commonly used significance levels. In all type I error plots, the dashed line corresponds to the Uniform(0,1) reference distribution expected under correct calibration.

Unless otherwise stated, each experiment uses $m = 5000$ simulation replications such that the Monte Carlo standard error (MCSE) for the empirical rejection rates $\hat{F}_{0.01}$, $\hat{F}_{0.05}$ and $\hat{F}_{0.10}$ are approximately 0.0014, 0.0031 and 0.0042 respectively. Approximate 95% Monte Carlo error bands, calculated as $\alpha \pm (1.96 \times \text{MCSE})$, are also presented on each plot as dotted lines. Furthermore, a thin plate regression spline is used with basis dimension $k = 10$ for each smooth across all simulations unless stated otherwise.

All simulations were carried out using R version 4.3.1, and ‘mgcv’ version 1.8-42. Representative R code for the simulations and the residual permutation comparison are provided in Appendices A–D, with the full simulation code available through the repository linked in Appendix E.

4.1 Simulation Scenario 1: Single null smooth

The first simulation scenario considers the simplest setting in which the true regression function does not contain a dependence on the covariate. This provides a baseline environment for assessing whether the approximate significance tests for smooth terms implemented in ‘mgcv’ are correctly calibrated when no nonlinear effect is present. By starting with this simple setting, we can isolate the effects of response family, sample size, smoothing parameter estimation method and basis dimension before moving on to more complex model structures.

For each simulation, a single covariate is generated by

$$x_i \sim \text{Uniform}(0, 1), \quad i = 1, \dots, n,$$

and a GAM is then fitted to each simulated dataset of the form

$$g(\mathbb{E}[y_i|x_i]) = c + f(x_i),$$

where y_i is the simulated response, g is the link function and f represents a smooth function. The response distributions, baseline parameter values and corresponding link functions are summarised in Table 1.

Response family	Response, y_i	Default parameters	Link, $g(x)$
Gaussian	$N(0, \sigma^2)$	$\sigma = 0.5$	x
Binomial	$\text{Bernoulli}(q)$	$q = 0.5$	$\text{logit}(x)$
Poisson	$\text{Poisson}(\lambda)$	$\lambda = 5$	$\log(x)$
Gamma	$\text{Gamma}(5, \mu/5)$	$\mu = 5$	$\log(x)$

Table 1: Response distributions, baseline parameter values and link functions used in Simulation Scenario 1.

The baseline parameter values were chosen to avoid extreme response behaviour and to ensure stable model fitting across response families. Under the DGP, the true smooth function satisfies the null hypothesis,

$$H_0 : f(x) = 0.$$

Any systematic departure of the resulting p -values from the $\text{Uniform}(0, 1)$ distribution therefore reflects miscalibration of the approximate significance test rather than genuine signal.

We vary several aspects of the model fitting procedure one at a time in order to assess its effect on type I error calibration. Unless stated otherwise, GCV smoothing parameter estimation method and sample size $n = 1000$ are used.

4.1.1 Effect of sample size

Across all four response families, sample size has a clear effect on type I error calibration in the baseline single null smooth setting. According to Table 2, the Gaussian and Gamma cases are mildly liberal at smaller sample sizes, with empirical rejection rates above nominal levels. This differs with the binomial case, which is conservative at smaller sample sizes but increases towards nominal with increasing sample size. It is interesting to note whilst the Gaussian and Gamma cases decrease toward nominal level, but never go

below it, the binomial case exceeds nominal level at the 5% and 10% levels at around a sample size of $n = 250$. The Poisson case is broadly well behaved but shows less systematic improvement with increasing n when compared to the Gaussian and Gamma cases.

	Gaussian		Binomial		Poisson		Gamma	
n	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
25	0.075	0.141	0.019	0.051	0.057	0.118	0.079	0.155
50	0.061	0.117	0.032	0.084	0.051	0.108	0.067	0.135
100	0.060	0.114	0.046	0.109	0.056	0.115	0.058	0.130
250	0.056	0.108	0.053	0.111	0.058	0.115	0.061	0.126
500	0.056	0.116	0.052	0.111	0.059	0.126	0.058	0.119
1000	0.052	0.113	0.060	0.125	0.051	0.109	0.055	0.113

Table 2: Empirical rejection rates at the 5% and 10% nominal levels for the null smooth $f(x)$ in Simulation Scenario 1 for varying sample size n .

Figure 1 shows more clearly that the Gaussian case is closest to nominal at the 5% nominal level in this scenario, with steady and stable improvement in calibration of p -values and successful calibration by about sample size 250. Gamma follows a similar trend, although calibration improves less clearly. The Poisson and binomial cases are more variable, but the binomial case in particular becomes better calibrated at larger sample sizes.

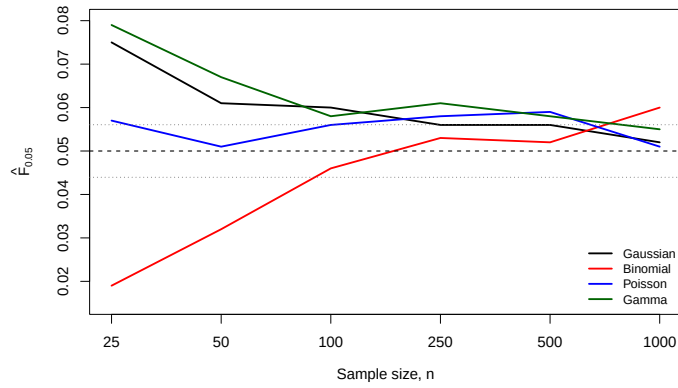


Figure 1: Empirical rejection rates at the 5% nominal level for the null smooth $f(x)$ in Simulation Scenario 1 for varying sample size n .

Similar conclusions are made for the smaller 1% nominal level, with calibration typically being more successful for all responses and for smaller

sample sizes.

n	Gaussian		Binomial		Poisson		Gamma	
	mean	sd	mean	sd	mean	sd	mean	sd
50	1.731	1.365	1.991	1.711	1.701	1.325	1.687	1.287
250	1.674	1.273	1.731	1.362	1.682	1.280	1.657	1.256
1000	1.682	1.297	1.734	1.335	1.683	1.259	1.629	1.213

Table 3: EDF summary for the null smooth $f(x)$ in Simulation Scenario 1 for varying sample size n .

The EDF summaries in Table 3 further assist the claim that increasing sample size improves inference in the simplest null case. For all four response families, the mean EDF is lower at moderate and large sample sizes than at $n = 50$, suggesting the model fitting attributes less complexity to the smooth which is closer to the true null. This effect is most evident in the binomial case, which has the largest EDF for smaller sample sizes, whilst the Gaussian, Poisson and Gamma cases decrease more moderately with increasing sample size.

4.1.2 GLM comparison

As an additional benchmark in the baseline null setting, we compare the approximate p -values for the smooth term $f(x)$ in the GAM with the p -values for the linear term in the GLM of the form

$$g(\mathbb{E}[y_i|x_i]) = c + \beta x_i,$$

using the same simulated datasets. Under the DGP, the x terms in both models are null. Since the significance tests in a GLM rely on less heuristic approximations, this comparison helps assess whether smooth term significance tests are well behaved compared to the corresponding standard parametric test applied under the same null DGP. Sample size $n = 500$ and REML smoothing parameter estimation method are used for this case.

Figure 2 shows that both the GLM linear term p -values and the GAM smooth term p -values remain close to nominal level across all response families. Within bounds of 95% Monte Carlo standard error, both of the empirical distributions are well calibrated, supporting the interpretation that the approximate smooth term test is reasonably reliable in the simplest null setting.

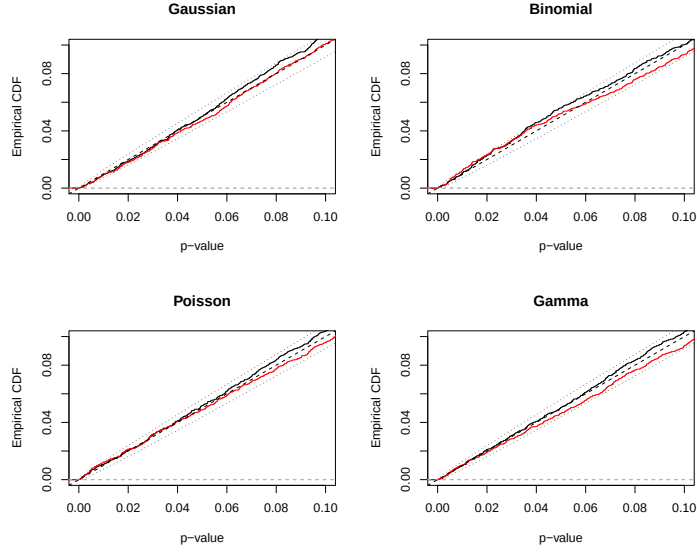


Figure 2: Empirical CDFs of p -values for Simulation Scenario 1 for the null smooth $f(x)$ in the GAM fit (black) and for the βx term in the GLM fit (red).

4.1.3 Effect of smoothing parameter estimation method

	Gaussian		Binomial		Poisson		Gamma	
Method	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
GCV	0.059	0.120	0.049	0.108	0.062	0.125	0.059	0.116
REML	0.056	0.113	0.047	0.104	0.056	0.115	0.055	0.111

Table 4: Empirical rejection rates at the 5% and 10% nominal levels for the null smooth $f(x)$ in Simulation Scenario 1 for varying smoothing parameter estimation method.

Across the four responses, Table 4 shows that the REML method consistently produces better calibrated p -values for the smooth term, in particular at larger nominal levels like 10%. This is most evident in the Poisson case, which reduces the empirical rejection rate at the 10% level from 0.125 for GCV to 0.115 for REML. Only in the binomial case does the method appear to have a less significant impact on significance test calibration, with there being less inflation from 5% nominal level for REML when compared to GCV.

The improvement is also clear in the EDF summary in Table 5, where

Method	Gaussian		Binomial		Poisson		Gamma	
	mean	sd	mean	sd	mean	sd	mean	sd
GCV	1.689	1.281	1.753	1.348	1.718	1.337	1.627	1.201
REML	1.294	0.580	1.294	0.564	1.296	0.574	1.299	0.578

Table 5: EDF summary for the null smooth $f(x)$ in Simulation Scenario 1 for varying smoothing parameter estimation method.

the mean and standard deviation of the EDF of the smooth term across all response distributions is much lower for the REML method rather than GCV. Since the EDF enters directly into the reference distribution for the smooth term test, these more stable EDF values may partly explain why the REML-based p -values are better calibrated than those obtained using GCV.

4.1.4 Effect of basis dimension

k	$\hat{F}_{0.01}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	EDF _{mean}	EDF _{sd}
5	0.010	0.049	0.105	1.256	0.474
10	0.009	0.046	0.102	1.295	0.572
20	0.009	0.046	0.101	1.306	0.606
30	0.009	0.046	0.101	1.307	0.612

Table 6: Empirical rejection rates and EDF summary for the null smooth $f(x)$ in Simulation Scenario 1 for varying basis dimension k for the Gaussian response.

Table 6 was simulated under sample size $n = 250$ and REML smoothing parameter estimation. The results indicate that basis dimension has little impact on type I error calibration in the baseline Gaussian null setting. As expected, the EDF summaries increase slightly with k , confirming the additional flexibility available to the smooth, but this does not impact the p -value calibration. Hence, provided it is already sufficiently large to capture flexibility in the data, basis dimension has very little impact when compared to sample size and smoothing parameter estimation method in this setting.

Overall, the baseline single null smooth setting suggests that type I error calibration depends primarily on sample size and smoothing parameter estimation method in this simple scenario, with response distribution having a smaller but still visible effect. Gaussian and Gamma responses show mild liberal behaviour at smaller sample sizes, whilst binomial response is initially

conservative before moving closer to nominal levels with increasing sample size. The Poisson response is comparatively stable, although its behaviour is less monotone. REML consistently provides more stable inference than GCV, and basis dimension has little effect once it is sufficiently large.

4.2 Simulation Scenario 2: Multiple null smooths

This simulation scenario extends the baseline null setting by fitting two null smooth terms simultaneously. This checks whether the approximate test remains well calibrated when another smooth term is present in the model, even when both covariates are generated independently and neither has a true effect. For each simulation, we generate two covariates independently by

$$x_i, z_i \sim \text{Uniform}(0, 1), \quad i = 1, \dots, n$$

and a GAM is fitted to each simulated dataset of the form

$$g(\mathbb{E}[y_i|x_i, z_i]) = c + f_1(x_i) + f_2(z_i),$$

with similar notation to Simulation Scenario 1 and where f_1 and f_2 both represent independent smooth functions. The same response generation process is carried over from Table 1. Under the DGP, each smooth function is null, so the hypotheses of interest are

$$H_{0,1} : f_1(x) = 0, \quad H_{0,2} : f_2(z) = 0.$$

We examine whether the presence of a second null smooth affects the calibration of smooth term p -values relative to Scenario 1. Unless stated otherwise, REML smoothing parameter estimation method and sample size $n = 500$ are used.

4.2.1 Effect of sample size

Table 7 shows that the empirical rejection rates for $f_1(x)$ and $f_2(z)$ are fairly similar across all four response families and sample sizes. There is only slight deviation between the calibration of the p -values between smooths for isolated cases, like binomial with sample size $n = 500$, which indicates that these are more dependent on the random nature in which the covariates and responses are generated. Relative to Scenario 1, adding another smooth

n	f_i	Gaussian		Binomial		Poisson		Gamma	
		$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
100	f_1	0.057	0.113	0.051	0.112	0.055	0.109	0.061	0.119
	f_2	0.060	0.118	0.051	0.103	0.053	0.103	0.056	0.116
250	f_1	0.058	0.111	0.051	0.113	0.051	0.113	0.057	0.113
	f_2	0.054	0.106	0.053	0.110	0.055	0.108	0.057	0.111
500	f_1	0.054	0.114	0.055	0.116	0.053	0.105	0.055	0.113
	f_2	0.049	0.103	0.048	0.107	0.053	0.109	0.053	0.113
1000	f_1	0.052	0.109	0.051	0.103	0.048	0.100	0.057	0.114
	f_2	0.055	0.109	0.052	0.108	0.053	0.108	0.052	0.111

Table 7: Empirical rejection rates at the 5% and 10% nominal levels for the null smooths $f_1(x)$ and $f_2(z)$ in Simulation Scenario 2 for varying sample size n .

where the covariate is generated similarly does not seem to introduce substantial additional distortion in the approximate significance tests. It is important to note that, opposed to what was seen in Scenario 1, the binomial case exhibits less conservatism in its p -values for smaller sample sizes and looks far more consistent with the other distribution families.

4.2.2 Effect of adding a second null smooth

To assess more directly whether the addition of a second null smooth affects the calibration of the test, we compare the empirical distribution of the p -values for the smooth $f_1(x)$ to the fitted smooth from Simulation Scenario 1, fitting both models to the same simulated datasets so that any differences reflect the change in model structure rather than variation in the generated data.

Figure 3 and Table 7 together suggest that the presence of a second independent null smooth does not substantially impact the calibration of the approximate significance tests. The empirical rejection rates for $f_1(x)$ and $f_2(z)$ remain near identical across response families according to the ECDF plots in Figure 3, indicating that GAM model expansion alone is not a key source of miscalibration in this simple independent null setting.

4.2.3 Effect of response parameter

For each response distribution, we vary the parameter that impacts the response variance to see if this has a substantial impact on type I error

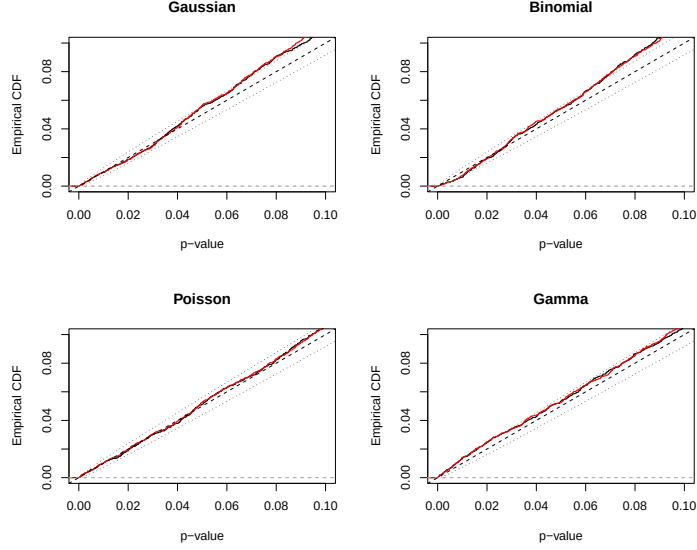


Figure 3: Empirical CDFs of p -values for the null smooth in the x covariate under GAM fits with one smooth term (black) and two smooth terms (red), generated relative to Simulation Scenario 1 and 2 respectively.

calibration in a simple null setting. Changing the variance structure of the response may affect the estimated covariance matrix and EDF used in the approximate smooth term test, making this a parameter of interest to explore. As seen previously, inference for the two nulls are similar and therefore, here we present the empirical rejection rates for only the smooth $f_1(x)$.

Figure 4 indicates that moderate change in the response parameter for each family has little systematic effect on empirical rejection rates for the Gaussian, Poisson and Gamma responses. Although fluctuation is visible, particularly in the Poisson case, there is no clear persistent pattern. In the binomial case, success probability $q = 0.5$ shows some improvement when compared to other values within the range considered. This is because $q = 0.5$ is the most balanced Bernoulli response, whilst other success probabilities are more likely to produce an imbalanced dataset, causing more extreme response values and poorer p -value calibration.

Overall, the addition of another null smooth does not have a substantial impact on the calibration of smooth term tests, and moderate variation in the parameters that generate the response has little systematic impact

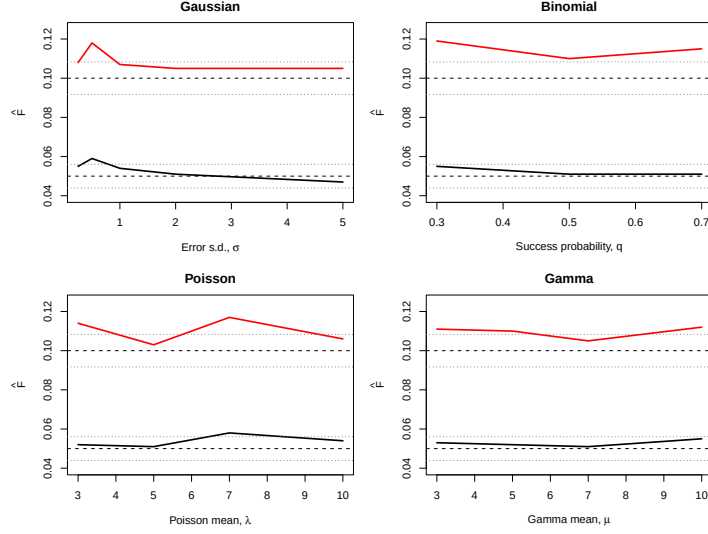


Figure 4: Empirical rejection rates at the 5% (black) and 10% (red) nominal levels for the null smooth $f_1(x)$ in Simulation Scenario 2 for varying response parameter.

across all responses, with slight improvement for $q = 0.5$ in the binomial response.

4.3 Simulation Scenario 3: Null smooth with correlated covariates

The third simulation scenario explores dependence between covariates. The aim is to assess whether the calibration of approximate smooth term significance tests deteriorates when a null smooth is fitted in the presence of a correlated linear covariate. For each simulation replication, we generate two covariates independently.

$$x_i \sim \text{Uniform}(0, 1), \quad u_i \sim N(0, 1), \quad i = 1, \dots, n.$$

Then, a nonlinear transformation of x_i is defined by $g(x_i) = \sin(2\pi x_i)$ and is standardised (represented by g^*). We generate the covariate $\tilde{z}_i$ as a linear combination of these u_i and $g^*(x_i)$, dependent on the correlation parameter ρ .

$$\tilde{z}_i = \rho g^*(x_i) + \sqrt{1 - \rho^2} u_i.$$

The covariate z_i used in the simulated dataset is the standardised version of $\tilde{z}_i$, ensuring that the results remain comparable across settings and are driven mainly by changes in the dependence structure. Finally, the response is generated according to Table 8.

Response family	Response, y_i	Mean structure
Gaussian	$z_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$	$\mathbb{E}[y_i z_i] = z_i$
Binomial	$\text{Bernoulli}(q_i)$	$\text{logit}(q_i) = \alpha z_i$
Poisson	$\text{Poisson}(\lambda_i)$	$\log(\lambda_i) = \log(5) + \alpha z_i$
Gamma	$\text{Gamma}(5, \mu/5)$	$\log(\mu_i) = \log(5) + \alpha z_i$

Table 8: Response generation and mean structure for each response distribution used in Simulation Scenario 3.

A GAM is then fitted of the form

$$g(\mathbb{E}[y_i|x_i, z_i]) = c + \beta z_i + f(x_i).$$

Under the DGP, the response depends on z_i only through a linear effect so the smooth is truly absent and the following hypothesis holds.

$$H_0 : f(x) = 0.$$

We investigate whether concurvity between covariates x and z cause any distortion in the approximate smooth term tests when the linear effect of z is correctly included in the model. Unless stated otherwise, REML smoothing parameter estimation method, $n = 500$, $\rho = 0.6$, $\alpha = 0.5$ and $\sigma = 0.5$ are used in simulations.

4.3.1 Effect of correlation parameter

ρ	Gaussian		Binomial		Poisson		Gamma	
	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
0.0	0.050	0.102	0.054	0.116	0.048	0.105	0.048	0.099
0.3	0.047	0.098	0.043	0.100	0.055	0.113	0.053	0.110
0.6	0.051	0.103	0.055	0.113	0.052	0.106	0.058	0.117
0.9	0.050	0.103	0.050	0.104	0.051	0.108	0.051	0.108

Table 9: Empirical rejection rates at the 5% and 10% nominal levels for the null smooth $f(x)$ in Simulation Scenario 3 for varying correlation parameter ρ .

Table 9 shows little evidence that increasing the correlation parameter ρ affects the calibration of the approximate significance test of the null smooth. Although fluctuations are visible in the binomial and Gamma cases in particular, there is no systematic trend. The Gaussian case is the most stable across all correlation parameter values considered, staying very close to nominal level with little fluctuation. Overall, these results suggest that dependence between covariates alone does not substantially distort the empirical rejection rates of the smooth term when the true linear effect is correctly included in the fitted model.

4.3.2 Effect of sample size

n	Gaussian		Binomial		Poisson		Gamma	
	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
50	0.054	0.116	0.037	0.093	0.050	0.106	0.068	0.129
100	0.055	0.112	0.046	0.101	0.050	0.101	0.058	0.120
250	0.053	0.106	0.046	0.101	0.053	0.110	0.053	0.113

Table 10: Empirical rejection rates at the 5% and 10% nominal levels for the null smooth $f(x)$ in Simulation Scenario 3 for varying sample size n .

Table 10 indicates that the approximate calibration of the significance test remains broadly comparable to the simpler null settings in the presence of moderate correlation between covariates. Increasing sample size has the most clear impact on improving calibration, with this being especially prevalent in the Gamma, Gaussian and binomial cases. This suggests that the inclusion of a correlated linear covariate is not by itself enough to produce significant miscalibration of type I error.

4.4 Simulation Scenario 4: Null smooths in the presence of a linear effect

The fourth simulation scenario extends Scenario 3 by adding a smooth term for the same covariate that already appears linearly in the model. This creates a more difficult setting for the approximate significance tests, since the GAM must distinguish between a genuine linear effect of z and any additional nonlinear contribution of the same covariate. The fitted GAM is of the form

$$g(\mathbb{E}[y_i|x_i, z_i]) = c + \beta z_i + f_1(x_i) + f_2(z_i),$$

where $f_2(z)$ is a smooth represented by a modified spline basis, so that the unpenalised linear component is removed from the smooth. Therefore, the linear trend in the GAM is entirely handled by the linear z_i term specified in the DGP, which results in two hypotheses of interest, both of which are true in this scenario.

$$H_{0,1} : f_1(x) = 0, \quad H_{0,2} : f_2(z) = 0.$$

We examine whether inference remains reliable when the model must distinguish the linear effect and the smooth term effect for a covariate, whilst also in the presence of dependence between covariates. The same standard parameters as with Scenario 3 apply here.

4.4.1 Effect of correlation parameter

ρ	Gaussian		Binomial		Poisson		Gamma	
	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
0.0	0.056	0.117	0.058	0.114	0.057	0.115	0.054	0.109
0.3	0.054	0.114	0.055	0.110	0.047	0.103	0.054	0.111
0.6	0.053	0.113	0.052	0.111	0.053	0.115	0.056	0.107
0.9	0.057	0.113	0.052	0.112	0.051	0.113	0.056	0.114

Table 11: Empirical rejection rates at the 5% and 10% nominal levels for the null smooth $f_1(x)$ in Simulation Scenario 4 for varying correlation parameter ρ .

Relative to Scenario 3, Table 11 shows that inference on the smooth term relating to the x covariate remains broadly similar for varying correlation parameter across all responses and is reasonably well calibrated, with some inflation at the 10% nominal levels as seen previously. Therefore, whilst varying correlation parameter ρ still has little impact on p -value calibration, we can also say the added complexity in the model has not significantly distorted inference for this smooth.

Table 12 shows that the approximate significance test for the null smooth $f_2(z)$ is substantially more liberal than for $f_1(x)$ across all four response families and correlation parameter values. This suggests that the additional smooth component of z is more prone to type I error inflation than the smooth of x , despite both being null under the DGP. However, as with $f_1(x)$, there is limited evidence that increasing ρ itself is responsible for

	Gaussian		Binomial		Poisson		Gamma	
ρ	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
0.0	0.063	0.120	0.068	0.122	0.065	0.120	0.055	0.109
0.3	0.067	0.128	0.065	0.128	0.060	0.116	0.063	0.117
0.6	0.070	0.122	0.062	0.117	0.066	0.123	0.067	0.125
0.9	0.064	0.114	0.063	0.124	0.060	0.121	0.068	0.129

Table 12: Empirical rejection rates at the 5% and 10% nominal levels for the null smooth $f_2(z)$ in Simulation Scenario 4 for varying correlation parameter ρ .

this inflation. The main feature of this scenario is instead the persistent difference between the smooth terms.

Taken together, these results suggest that the difficulty in this scenario lies in separating a true linear effect from an additional null smooth with the linear component removed from its basis.

4.4.2 Effect of sample size

		Gaussian		Binomial		Poisson		Gamma	
n	f_i	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
100	f_1	0.062	0.116	0.044	0.105	0.052	0.108	0.059	0.121
	f_2	0.063	0.117	0.065	0.134	0.064	0.121	0.069	0.117
250	f_1	0.052	0.110	0.055	0.116	0.054	0.119	0.057	0.108
	f_2	0.066	0.125	0.067	0.129	0.065	0.122	0.070	0.127
500	f_1	0.057	0.115	0.059	0.121	0.053	0.108	0.053	0.109
	f_2	0.063	0.115	0.063	0.123	0.065	0.122	0.063	0.113
1000	f_1	0.055	0.108	0.050	0.111	0.053	0.106	0.051	0.106
	f_2	0.062	0.114	0.065	0.124	0.062	0.120	0.065	0.123

Table 13: Empirical rejection rates at the 5% and 10% nominal levels for the null smooths $f_1(x)$ and $f_2(z)$ in Simulation Scenario 4 for varying sample size n .

Table 13 indicates that the inference of the smooth $f_1(x)$ remains similar to those seen in previous scenarios, and that calibration of this smooth improves with sample size in this setting. By contrast, the approximate significance tests for the smooth $f_2(z)$ are consistently more liberal than that for $f_1(x)$ for all response families. This gap persists as sample size increases and there is very little evidence that the approximate p -values improve with increasing sample size. This indicates that this poor calibration

in the smooth separating the linear and nonlinear components is not simply a small-sample phenomenon and is unlikely to be primarily caused by the asymptotic normality assumption in the construction of the reference distribution.

4.4.3 Effect of smoothing parameter estimation method

Since Table 4 in Simulation Scenario 1 indicated little difference between GCV and REML methods for p -value calibration in the binomial case, we return to varying this parameter in the more complicated Simulation Scenario 4.

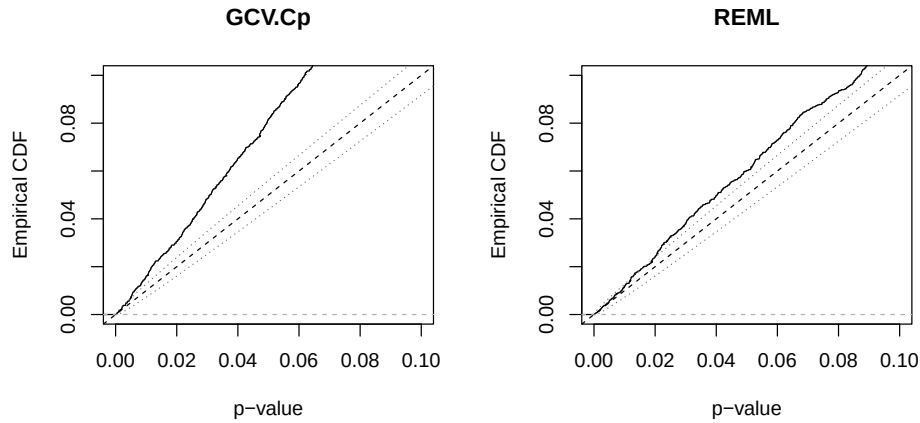


Figure 5: Empirical CDFs of p -values for the null smooth $f_2(z)$ in Simulation Scenario 4 in the binomial case for varying smoothing parameter estimation method.

Figure 5 gives clear indication that REML provides empirical rejection rates much closer to the nominal level rather than GCV, which approximates much more liberal p -values consistently across the range considered. Whilst the approximated p -values from REML are moderately liberal, it is much less than in the GCV case, and thus REML provides more reliable inference than GCV. This suggests that the choice of smoothing parameter estimation method becomes more important in the binomial case for more complex GAMs.

4.4.4 GLM comparison

To clarify the source of the poor calibration observed for the null smooth $f_2(z)$, we compare the fitted GAM with the corresponding correctly specified GLM on the same simulated datasets. Under the DGP in this scenario, the response depends on z on linear effect only, so the correctly specified GLM is of the form

$$g(\mathbb{E}[y_i|x_i, z_i]) = c + \beta z_i.$$

This GLM is fitted as a benchmark for the parametric part of the model, rather than as a competitor to the GAM. The comparison helps to determine whether the poor calibration of the smooth $f_2(z)$ is due to the data generating setting itself, or instead arises specifically from the GAM's attempt to separate linear and nonlinear components of a covariate. If the GLM and GAM both estimate β accurately, but the GAM still produces inflated rejection rates for $f_2(z)$, then the poor calibration cannot be explained by failure to identify the linear effect.

	Gaussian		Binomial		Poisson		Gamma	
Model	$\hat{\beta}_{\text{mean}}$	$\hat{\beta}_{\text{sd}}$	$\hat{\beta}_{\text{mean}}$	$\hat{\beta}_{\text{sd}}$	$\hat{\beta}_{\text{mean}}$	$\hat{\beta}_{\text{sd}}$	$\hat{\beta}_{\text{mean}}$	$\hat{\beta}_{\text{sd}}$
GAM	1.000	0.044	0.509	0.195	0.501	0.038	0.499	0.040
GLM	1.000	0.022	0.505	0.098	0.500	0.019	0.500	0.020

Table 14: Mean and standard deviation of the estimated linear coefficient β across simulations for Simulation Scenario 4 for the specified GLM and GAM models.

	Gaussian		Binomial		Poisson		Gamma	
Model	$\hat{F}_{0.01}$	$\hat{F}_{0.05}$	$\hat{F}_{0.01}$	$\hat{F}_{0.05}$	$\hat{F}_{0.01}$	$\hat{F}_{0.05}$	$\hat{F}_{0.01}$	$\hat{F}_{0.05}$
GAM	1.000	1.000	0.744	0.828	0.997	0.998	0.994	0.996
GLM	1.000	1.000	0.998	1.000	1.000	1.000	1.000	1.000

Table 15: Empirical rejection rates at the 1% and 5% nominal levels for the linear term βz in Simulation Scenario 4 for the specified GLM and GAM models.

Table 14 indicates that the true linear effect is being recovered accurately in the GAM and there is not much difference in the mean across simulations in the estimated coefficient when compared to the GLM across all response families. The presence of the smooth terms does make the inference of the linear term less precise since the estimated coefficient has more variability,

particularly in the binomial case. This is especially clear in the empirical rejection rates in Table 15, where the rejection rates of the true linear term are much lower for the binomial GAM than for the GLM. Nevertheless, across all response families, the linear term is strongly detected in both models and is recovered accurately. This suggests that the poor calibration in the null smooth $f_2(z)$ is not primarily due to a failure to identify the linear contribution of z . Instead, the main difficulty in Simulation Scenario 4 appears to lie in distinguishing the true linear contribution from an additional smooth component of the same covariate that is null under the DGP.

4.4.5 Residual permutation comparison

Since this scenario shows persistent inflation when separating linear and nonlinear contributions of the same covariate, an additional permutation-based test was implemented. The aim is to assess whether this behaviour is mainly caused by the approximate reference distribution used by ‘mgcv’, or whether it persists under an empirical null distribution and therefore reflects a broader difficulty in separating the two components.

Permutation tests provide a resampling based alternative to the approximate reference distributions considered here by constructing an empirical null distribution for a test statistic through repeated rearrangement of the data under the null hypothesis. Following the framework described by Young et al. (2011), significance can be assessed by repeatedly permuting the covariate associated with the smooth term while preserving relationships between the response and other predictors, refitting the GAM to each permuted dataset, and comparing the observed statistic with the resulting permutation distribution. They show that reselecting the smoothing parameters for each permuted dataset provides more reliable inference when compared to using the same smoothing parameters from the original fit before permuting the dataset. Although the present study uses penalised GAMs in ‘mgcv’ rather than LOESS GAMs, the same principle is relevant, and we use the method of reselecting smoothing parameters for each permuted dataset. The null hypothesis is rejected when the observed statistic in the first fitting lies in the upper tail of this empirical distribution. Their study finds that permutation-based tests can provide more reliable type I error control with respect to deviance test statistics.

Motivated by their study, the work from Young et al. is adapted to the

approximate F -statistic reported by ‘mgcv’ for the smooth term, allowing comparison between approximate p -values and permutation based p -values within the Scenario 4 framework. The permutation approach they use is unsuitable here as the z covariate has two terms included in the model, and would make the smooth term depend on a different, artificially permuted version of z from the one used in the linear term, effectively creating an additional covariate not present under the null model. Therefore, we implement a residual permutation test procedure, following a process similar to the Freedman-Lane method as outlined by Winkler et al. (2014).

The residual permutation test is only implemented for the Gaussian response setting in Simulation Scenario 4. This restriction is made due to permutation inference relying on exchangeability of observations under the null hypothesis (Good; 2005, p. 268), which holds for the Gaussian GAM but is not guaranteed for binomial, Poisson or Gamma responses where the variance depends on the mean. In such cases, more specialised resampling schemes are required. This restriction was also made due to the intensive computational runtime of the procedure.

For each simulated dataset, the full GAM was first fitted,

$$\mathbb{E}[y_i|x_i, z_i] = c + \beta z_i + f_1(x_i) + f_2(z_i),$$

and the reported ‘mgcv’ p -value and corresponding F -statistic for the smooth term $f_2(z)$ were extracted. This observed statistic is denoted F_{obs} . To generate the empirical null distribution, the null model

$$\mathbb{E}[y_i|x_i, z_i] = c + \beta z_i + f_1(x_i)$$

was then fitted, removing the smooth term $f_2(z)$ from the full GAM. Let $\hat{y}_i$ and $e_i = y_i - \hat{y}_i$ denote the fitted value and residual respectively from this null model for observation i . Then, for each permutation $b = 1, \dots, 499$, we generate pseudo-response by

$$y_i^{(b)} = \hat{y}_i + e_{\pi_b(i)},$$

where π_b is a random permutation of the residual indices. The full GAM was then refitted to each pseudo-response $y_i^{(b)}$, with smoothing parameters re-estimated with each fit, giving permutation statistics $F_{\text{perm}}^{(b)}$, $b = 1, \dots, 499$.

This is then compared to the F -statistic obtained from the original fit by the following formulae.

$$p_{\text{perm}} = \frac{1 + \sum_{b=1}^{499} \mathbf{1}[F_{\text{perm}}^{(b)} \geq F_{\text{obs}}]}{500}.$$

The use of this form of permutation p -value follows Phipson and Smyth (2010), who recommend this corrected expression. This is then repeated $m = 5000$ times and the p -values under permutation tests are treated in a similar way to the observed p -values from ‘mgcv’. The results are shown in Table 16 and Figure 6, where the permutation ECDF appears as a step function because, with 499 permutations, the smallest possible change in the permutation p -value is $1/500 = 0.002$.

Model	$\hat{F}_{0.01}$	$\hat{F}_{0.05}$	$\hat{F}_{0.10}$
mgcv F -test	0.013	0.064	0.121
Permutation F -test	0.010	0.053	0.099

Table 16: Empirical rejection rates at the 1%, 5% and 10% nominal levels for the smooth term $f_2(z)$ in Simulation Scenario 4 in the Gaussian case, comparing the approximate ‘mgcv’ F -test with the residual permutation F -test.

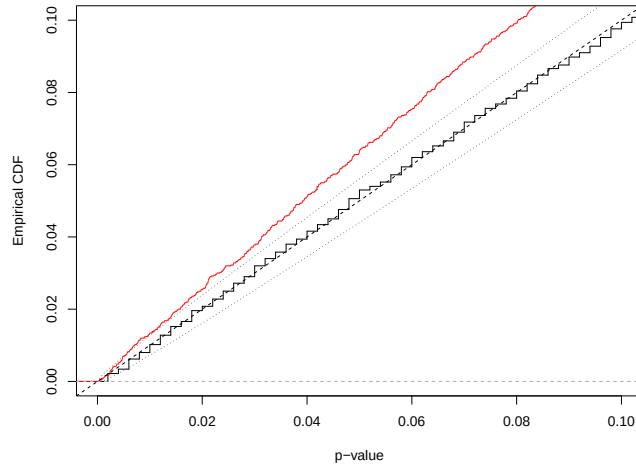


Figure 6: Empirical CDFs of p -values for the null smooth $f_2(z)$ in Simulation Scenario 4 in the Gaussian case under ‘mgcv’ inference (red) and residual permutation test inference (black).

The residual permutation based test gives empirical rejection rates much closer to nominal levels than the ‘mgcv’ approximate F -test. At the 1%, 5% and 10% nominal levels, the residual permutation rejection rates are 0.010, 0.053 and 0.099 respectively, all lying within the corresponding 95% MCSE bands shown in Figure 6. By comparison, the approximate ‘mgcv’ test gives rejection rates of 0.013, 0.064 and 0.121. This indicates that the residual permutation procedure substantially reduces the inflation observed for the more difficult null smooth $f_2(z)$, particularly at the 5% and 10% levels. Therefore, in this Gaussian setting, a large part of the miscalibration observed appears to come from the approximate reference distribution used by ‘mgcv’, rather than from model structure alone. However, this conclusion should be interpreted cautiously. The residual permutation comparison is only implemented in the Gaussian case, where the exchangeability assumptions are more straightforward and cannot be assumed to carry over to other responses. In addition, the procedure is substantially more computationally expensive than the approximate ‘mgcv’ test, which is an important consideration once working with large datasets.

4.5 Summary of type I error study

Overall, the type I error study suggests that the approximate significance tests for smooth terms in ‘mgcv’ are reasonably well calibrated in simpler settings, although some moderate inflation remains at larger nominal levels and in smaller samples. Across the scenarios considered, REML generally provides more reliable inference than GCV, whilst basis dimension has little effect in the simple null setting explored here. The results also indicate that correlation between covariates alone is not sufficient to cause substantial distortion in the approximate smooth term tests.

The main difficulty instead arises in the more complex setting where the model must distinguish between a true linear effect and an additional null smooth of the same covariate. In this setting, the approximate p -values for the null smooth remain noticeably liberal across all response families. The GLM comparison suggests this is not primarily due to the failure to recover the true linear effect. The Gaussian residual permutation comparison substantially reduced the inflation observed, with empirical rejection rates much closer to nominal levels than the approximate ‘mgcv’ test. This suggests a substantial part of the miscalibration is due to the construction

of the reference distribution in ‘mgcv’ in this Gaussian response case.

The next chapter therefore examines whether this same challenging setting also leads to a deterioration in power when a true nonlinear effect is present.

5 Power Assessment Study

Having examined the type I error behaviour of the approximate significance tests for smooth terms, we now consider their ability to detect genuine smooth effects when the null hypothesis is false. In this study, the aim is not to compare to the Uniform(0,1) distribution. Instead, a powerful test should produce p -values concentrated closer to zero. We therefore examine the empirical distribution of p -values over the full range $p \in [0, 1]$, where a greater concentration near zero appears as a shift towards the top left of the ECDF plots. The aim of this chapter is to assess whether the approximate tests respond as expected to increasing sample size and signal strength, and whether the difficulties in the type I error study also affect detection of genuine nonlinear effects in the corresponding power scenario.

5.1 Simulation Scenario 1: Single smooth under a false null

This simulation is analogous to the first simulation scenario for the type I error study, where we consider a simple baseline setting for assessing power of the approximate smooth term significance tests. We use the same covariate generation and fitted GAM structure, but with responses now generated under a false null so that the smooth term is genuinely nonzero. This is shown in Table 17, where δ is the signal strength parameter and h is the true smooth function.

Response family	Response, y_i	Mean structure
Gaussian	$\delta h(x_i) + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$	$E[y_i x_i] = \delta h(x_i)$
Binomial	Bernoulli(q_i)	$\text{logit}(q_i) = \delta h(x_i)$
Poisson	Poisson(λ_i)	$\log(\lambda_i) = \log(5) + \delta h(x_i)$
Gamma	Gamma(5, $\mu/5$)	$\log(\mu_i) = \log(5) + \delta h(x_i)$

Table 17: Response generation and mean structure for each response distribution used in Power Simulation Scenario 1.

Unless specified otherwise, we take $h(x)$ to be a standardised version of $\sin(2\pi x)$. We also use $n = 500$, REML smoothing parameter estimation method, and δ as 0.05, 0.20, 0.04 and 0.04 for the Gaussian, binomial, Poisson and Gamma responses respectively across simulations. These standard δ values were chosen to give broadly comparable baseline detectability at the smallest sample size considered. This allows the effect of increasing sample size to be compared across response families without one response

dominating the plot because its signal was much stronger on the scale of the response.

5.1.1 Effect of sample size

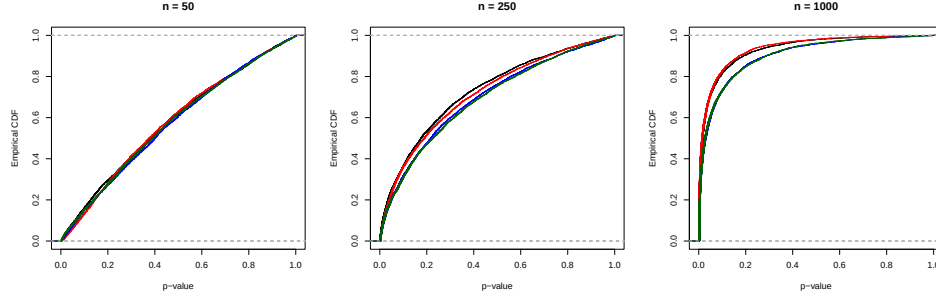


Figure 7: Empirical CDFs of p -values for the true smooth $f(x)$ in Power Simulation Scenario 1 for varying sample size and response family, where black denotes Gaussian, red denotes binomial, blue denotes Poisson and green denotes Gamma.

According to Figure 7, for all response families, increasing sample size shifts the empirical distribution of p -values further to the top left, indicating a growing concentration of small p -values across simulations and hence, increasing power to detect the nonzero smooth. All responses are comparable in their rate of improvement with sample size.

5.1.2 Effect of true smooth function

In this case, both true smooth functions were standardised before being multiplied by δ to ensure the comparison is driven by shape, rather than having a larger scale. For each response family, Figure 8 shows that the empirical distribution of p -values for the cubic signal lies above that for the sinusoidal signal, which could suggest that under these conditions, the cubic effect is marginally easier to detect. However, the difference between the two curves is not large, so the overall conclusion is that the approximate tests have comparable power for both smooth functions considered.

5.1.3 Effect of signal strength parameter

The results for varying signal strength parameter were generated with $\delta \in \{0.10, 0.25, 0.50, 1.00\}$ for the binomial case and $\delta \in \{0.01, 0.05, 0.10, 0.20\}$

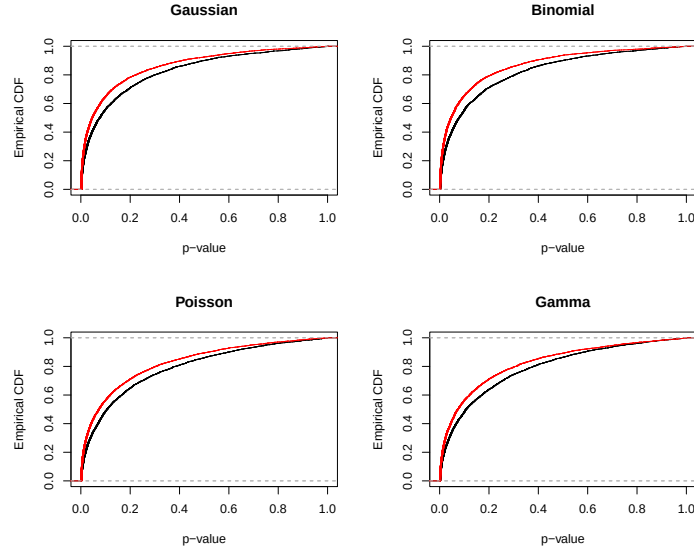


Figure 8: Empirical CDFs of p -values for the true smooth $f(x)$ in Power Simulation Scenario 1 for varying response family and standardised true smooth function $h(x)$, where black denotes $\sin(2\pi x)$ and red denotes $(x - 0.5)^3$.

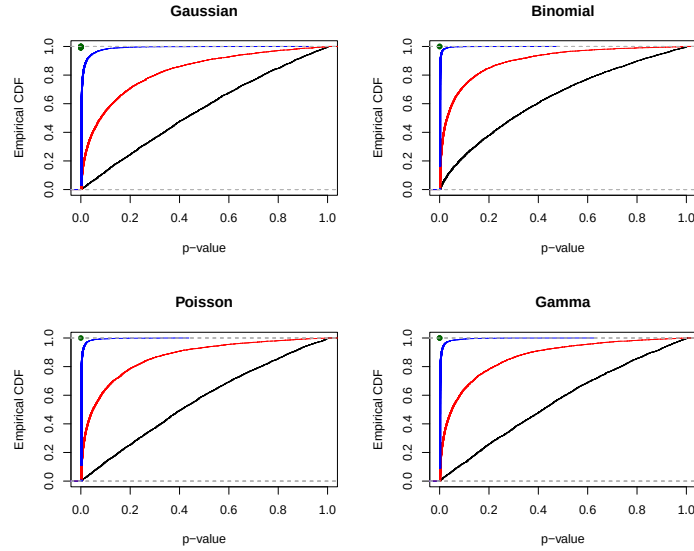


Figure 9: Empirical CDFs of p -values for the true smooth $f(x)$ in Power Simulation Scenario 1 for varying response family and signal strength parameter δ , where black, red, blue and green denote smallest to largest parameter respectively within each response family.

otherwise. A larger range of δ values is used for the binomial case since the effect is applied on the logit scale, so observations contain less information per observation when compared to the Gaussian case, for example. The plots therefore show the transition of weak to strong power for each family internally, rather than being directly comparable across response distributions. It is clear across all responses from Figure 9 that stronger nonlinear signals are detected with greater reliability in the baseline null case, which is the behaviour expected from a powerful test and provides good indication that the approximate tests perform reliably in the baseline power scenario.

5.2 Simulation Scenario 2: Detecting nonlinearity beyond a linear effect

The second power simulation scenario is motivated by the most difficult setting identified in the type I error study where the fitted GAM must separate a linear effect from an additional smooth contribution for the same covariate. We examine whether, in the corresponding false null setting, the approximate significance test is able to detect a genuine nonlinear departure from linearity. We use the same covariate generation and fitted GAM structure from the type I error Simulation Scenario 4, whilst the response generation is changed and shown in Table 18.

Response family	Response, y_i	Mean structure
Gaussian	$z_i + \delta h(z_i) + \epsilon_i$	$\mathbb{E}[y_i x_i, z_i] = z_i + \delta h(z_i)$
Binomial	$\text{Bernoulli}(q_i)$	$\text{logit}(q_i) = \alpha z_i + \delta h(z_i)$
Poisson	$\text{Poisson}(\lambda_i)$	$\log(\lambda_i) = \log(5) + \alpha z_i + \delta h(z_i)$
Gamma	$\text{Gamma}(5, \mu/5)$	$\log(\mu_i) = \log(5) + \alpha z_i + \delta h(z_i)$

Table 18: Response generation and mean structure for each response distribution used in Power Simulation Scenario 2.

In this case, the true hypotheses are

$$H_{0,1} : f_1(x) = 0, \quad H_{1,2} : f_2(z) \neq 0.$$

Unless stated otherwise, $h(z)$ is a standardised version of $z^2 - 1$, with REML smoothing parameter estimation method, sample size $n = 500$, $\delta = 0.20$ in the binomial case and $\delta = 0.02$ for the other responses are used in this scenario. As in the first power simulation, the δ values were chosen to show a clear transition from weak to strong power within each response family,

rather than to represent directly comparable effect sizes across families. The same baseline choices are made as in the corresponding type I error scenario.

5.2.1 Effect of signal strength relative to null rejection rate

We compare the empirical rejection rates for the true smooth $f_2(z)$ with the corresponding rejection rates for the null smooth under the null hypothesis in type I error Simulation Scenario 4, using the same fitted model structure. For each simulation setting, two responses are generated using the same covariates and fitted with the same GAM structure; one response is generated under the corresponding null setting, and the other under the alternative with nonlinear signal strength δ . Comparing these quantities shows whether the rejection rate for $f_2(z)$ increases beyond its already inflated null level as the nonlinear signal becomes stronger. These results are shown in Table 19, along with the empirical CDF plots for increasing signal strength in Figure 10, where the plots use $\delta \in \{0.05, 0.10, 0.20, 0.30\}$ in the binomial case and $\delta \in \{0.005, 0.01, 0.02, 0.04\}$ for the other responses.

Response	δ	$\hat{F}_{0.05}^{\text{null}}$	$\hat{F}_{0.05}^{\text{alt}}$	$\hat{F}_{0.05}^{\text{alt}} - \hat{F}_{0.05}^{\text{null}}$
Gaussian	0.01	0.069	0.084	0.015
	0.04	0.069	0.414	0.355
Binomial	0.10	0.059	0.171	0.112
	0.30	0.064	0.831	0.767
Poisson	0.01	0.066	0.092	0.026
	0.04	0.061	0.548	0.487
Gamma	0.01	0.071	0.088	0.017
	0.04	0.062	0.488	0.426

Table 19: Empirical rejection rates at the 5% nominal level for the smooth term $f_2(z)$ under the corresponding null setting and alternative setting with increasing signal strength δ .

Table 19 and Figure 10 together show that the rejection rate for $f_2(z)$ increases as signal strength increases, with the empirical CDF shifting towards the top left side of the plot. For small values of δ , the alternative rejection rates are only slightly above the corresponding null rejection rates, reflecting the difficulty in detecting a weak nonlinear departure from linearity. There is a substantial increase in the difference between these empirical rejection rates for larger values of δ , indicating that when the nonlinear signal is sufficiently strong, the smooth term test is able to detect genuine departures

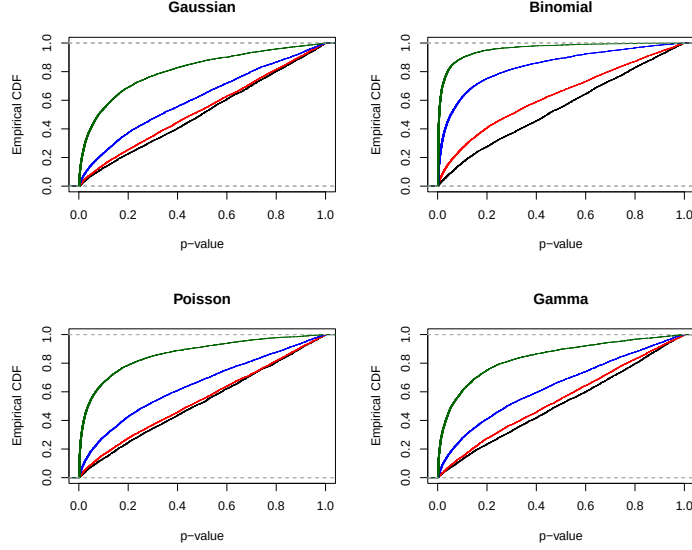


Figure 10: Empirical CDFs of p -values for the true smooth $f_2(z)$ in Power Simulation Scenario 2 for varying response family and signal strength parameter, where black, red, blue and green denote smallest to largest δ for each response respectively within each response family.

from linearity. However, since the null rejection rates for $f_2(z)$ are already inflated, the alternative rejection rates should be interpreted relative to this inflated baseline rather than as evidence of a powerful test.

5.2.2 Effect of correlation parameter

Similar to the results from the type I error study, Figure 11 shows that the correlation parameter has very little substantial impact on the power of a true smooth. This reinforces that correlation structure between x and z is not a key source of difficulty for this scenario.

5.3 Summary of power study

The power study indicates that the approximate significance tests in ‘mgcv’ are effective at detecting genuine nonlinear effects in relatively simple settings. The baseline single smooth scenario showed that increasing sample size and signal strength both moved the empirical distribution of p -values towards zero across all response families. The comparison of different true smooth functions suggests that the precise shape of the power curve may

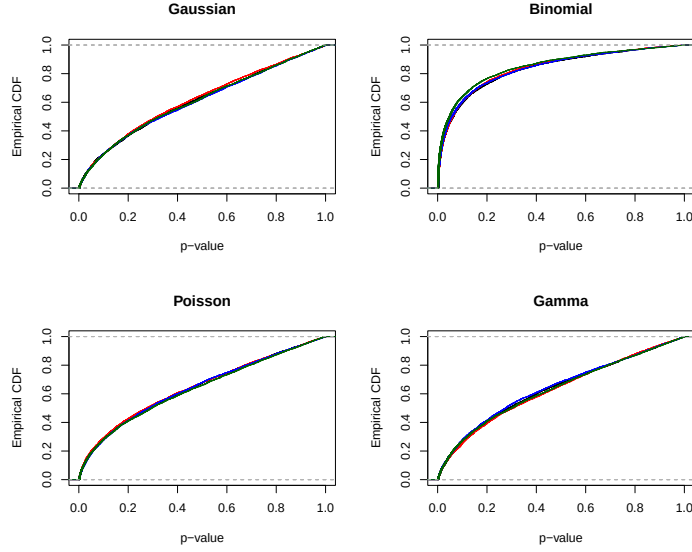


Figure 11: Empirical CDFs of p -values for the true smooth $f_2(z)$ in Power Simulation Scenario 2 for varying response family and correlation parameter, where black, red, blue and green denote smallest to largest correlation parameter, $\rho \in \{0, 0.3, 0.6, 0.9\}$ respectively.

change, but both of the signals considered were detected reasonably well.

In the more challenging linear and smooth setting, the approximate significance test for the smooth with its linear term in its basis removed was shown to be effective at detecting genuine nonlinear departures from linearity, provided the signal strength was large enough. However, this must be interpreted with the corresponding type I error of the same smooth being inflated under the null, so this power may be harder to distinguish in real applications. Correlation parameter also had very little impact on the power in this study.

The next chapter discusses the findings from both studies together, considering their implications for the practical use of GAM significance tests and outlining the main limitations of the study.

6 Discussion

The empirical simulation studies have investigated the reliability of the approximate significance tests for smooth terms implemented in the ‘mgcv’ framework. This chapter contextualises them within the theory developed from Chapter 3, compares findings with existing literature, explains the limitations of the studies, and suggests where further study could be conducted.

Reliability of approximate smooth term tests in simple settings

In simple null settings, the approximate p -values are generally well calibrated, with some inflation remaining at larger nominal levels in the range $p \leq 0.10$. This observation coincides with Wood (2013), who proposed the Wald-type smooth term test implemented in ‘mgcv’ and showed that the resulting p -values provide reasonably successful null behaviour through assessment of their empirical behaviour. The contribution of the present study has expanded Wood’s comparison by examining the behaviour across more response families, smoothing estimation methods and covariate correlation settings, providing further insight beyond Wood’s study into which modelling features cause the approximate construction of the significance tests to break down or remain reliable.

Increasing sample size had a very noticeable improvement on calibration, particularly for the Gaussian and Gamma responses, whilst the Poisson case was comparatively stable at smaller sample sizes. The binomial case was less consistent, showing both conservative and mildly liberal behaviour in the approximate significance tests dependent on the setting considered, but still generally showing improvement in calibration with sample size. The power scenarios mirrored this improvement, where increasing sample size or the signal for the true smooth led to a greater concentration of p -values near zero. The improvement in calibration of type I error and power seen with increasing sample size coincides with the approximate normality of the estimated coefficient vector used in the significance tests, which holds for all responses given a sufficiently large sample size. The binomial case is likely to provide less stable inference since binary data contains less information per observation, and imbalance in the response can make the large sample approximation and fitted covariance structure more sensitive to the simulated data. Nevertheless, the overall effect of response family in the simple

scenarios was secondary to the effects of sample size.

One of the clearest findings across the null simulation scenarios was that REML gave better calibrated smooth term p -values than GCV. Across all four responses, REML generally produced smaller empirical rejection rates closer to the nominal level, whilst also providing more stable EDF summaries for each null smooth. This behaviour is consistent with the findings of Wood (2011), who showed that the GCV score is more likely to produce multiple minima with respect to the smoothing parameter, which can lead to more unstable smoothing parameter estimates and undersmoothing. By contrast, REML tends to penalise overfitting more severely and gives a more pronounced optimum. This difference in methods is particularly relevant for inference because the smooth term tests treat the estimated smoothing parameters as fixed when constructing the covariance matrix and EDF used in the reference distribution. If smoothing parameter selection is unstable, this additional uncertainty is not fully reflected in the reported p -values. The simulation results suggest that REML reduces this source of instability, implying that it is the preferable smoothing parameter estimation method over GCV in simple settings when the main aim is smooth term inference rather than model fitting.

By contrast, increasing the number of independent null smooths, varying response parameters, varying basis dimension of the fitted smooth and introducing correlation between different covariates included in the model had very little impact on p -value inference in simpler null settings. This is consistent with the construction of the approximate tests as the changes in the parameters discussed would be unlikely to substantially disrupt the normal approximation, the estimated covariance structure, or the EDF used in the reference distribution, particularly in the simple null settings explored. The limited effect of the basis dimension on inference is also consistent with the work by Wood discussed in Chapter 2, who states that once the basis dimension is sufficiently flexible, it is not a key source of concern in model fitting. Similarly, the shape of the true smooth also did not massively affect the power of the approximate significance tests in the baseline power scenario. This suggests that both functions were well represented by the spline basis and that signal strength and sample size were more important determinants of detection, coinciding with sample size being the more important factor affecting the approximations in the significance tests.

Difficulty separating linear and nonlinear effects

The clearest deterioration in the approximate smooth term tests occurred when the fitted GAM was required to separate a true linear effect from an additional smooth term for the same covariate with the linear term removed from its smooth basis. The type I error study showed noticeably greater inflation over simpler null scenarios, and this inflation persisted across all responses and sample sizes considered. Since increasing sample size did not remove the inflation, the deterioration does not appear to be primarily a small sample failure of the asymptotic normality approximation.

Further comparison showed that the linear term was recovered accurately and similarly to the corresponding simpler GLM model, suggesting that the observed poor calibration was not primarily due to a failure to identify the linear contribution of the covariate. Therefore, the difficulty appears to arise more specifically from testing whether there is any additional nonlinear structure in a covariate whose true linear contribution has already been included.

This distinction is important as the smooth term in this scenario is not being tested as a general covariate effect. Instead, it is a more restrictive test purely for testing departure from linearity. Under the null hypothesis, any remaining curvature in the covariate is random variation rather than genuine signal. However, the restricted smooth basis may still be able to represent small random departures from linearity, producing a nonzero EDF and an apparently meaningful smooth component. The approximate F -distribution may not take this additional uncertainty into account, which causes the resulting test to be too liberal under the null.

This interpretation is related to the inferential difficulties discussed by Marra and Wood (2012). They show that Bayesian confidence intervals for the smooth terms of a GAM can behave poorly when the true effect of a covariate lies closely to the null space of the smoothing penalty. Although their work considers confidence interval coverage rather than p -value calibration, the present setting has a similar structure. The true effect is linear, while the test only concerns whether any remaining nonlinear component is needed. The observed inflation therefore suggests that the altered smooth basis and the approximate reference distribution together create a more difficult inferential problem than the simple zero effect tests considered earlier.

The residual permutation comparison supports the interpretation that

the approximate reference distribution itself is a cause of inflation in the Gaussian case. Replacing the approximate ‘mgcv’ reference distribution with an empirical null distribution substantially reduced the inflation, highlighting that alternative inferential procedures can improve calibration in this stressed setting. This specifically implies that the approximate F -distribution itself contributes meaningfully to the inflation observed in this setting for Gaussian responses. Clearly, together, the model structure creates a difficult testing problem, and the approximate reference distribution does not fully calibrate the uncertainty introduced by that problem. However, the permutation test should not be considered a replacement for the ‘mgcv’ significance tests. In particular, they are much more computationally expensive and the test explored here was limited to the Gaussian case and may not produce similar improvements for other responses. Here, they are considered key evidence of the source of inflation and motivate further study of alternative inferential procedures.

The corresponding power study implied that under the alternative hypothesis, this additional smooth was able to detect genuine departures from linearity, provided that the strength of the nonlinear signal was large enough. Since the null rejection rates were already inflated, the power of a test for a smooth of this form should be interpreted with caution, particularly if the nonlinear signal to be distinguished is likely to be weak.

In practical applications, this means that a small ‘mgcv’ p -value for a smooth term should be interpreted more cautiously when the smooth is being used to test for nonlinearity beyond an already included linear effect. In such cases, the p -value may indicate some evidence of curvature, but it should not be treated as exact evidence of a nonlinear effect without additional diagnostic checks or alternative inferential procedures.

Limitations of the simulation study

The main strength of the simulations used here is that they isolate specific modelling features, allowing changes in calibration to be attributed more clearly to changes in parameters or model structure. However, this control necessarily comes at the cost of simplifying some features of applied GAM settings, and so, the conclusions of this study should be interpreted in the context of several limitations in the simulation design.

One limitation concerns the simplicity of the covariate structures used

in the simulations. Across the study, covariates were generated from either $U(0, 1)$ or $N(0, 1)$ distributions, which allowed for controlled comparisons across response families and ensured stable model fitting. However, applied datasets often involve more irregular covariate distributions, including skewness, outliers, or clustering. Since smoothing parameter estimation depends directly on the observed data, and in turn affects the covariate matrix used by the significance tests, these features could affect the calibration of smooth term tests in ways not captured by the present simulations. Similarly, the covariate correlation settings considered here were relatively simple and low dimensional. Although this allowed the effect of pairwise correlation to be isolated, applied GAMs may involve more complex concavity between several covariates.

A related limitation concerns the choice of signal strength parameters in the power simulations. The values of δ were chosen to show a clear transition from weak to strong power within each response family, rather than to define directly comparable effect sizes between responses. Therefore, the power results were mainly restricted to interpretation within each response family, rather than as direct comparisons of power between responses.

Another limitation is that the simulations considered independent data only. The results therefore apply most directly to standard GAM settings, but do not consider residual correlation or grouped random effects. In many applications, observations may be dependent in some way, which is likely to affect the covariance matrix used for significance testing and distort p -value inference.

The simulations also focused on relatively simple additive models with univariate smooths. This limits how far the conclusions can be generalised to GAMs involving interaction effects or higher dimensional smooth terms. Tensor product smooths, for example, involve separate penalties for smoothing parameters for different covariates, as explained by Wood (2006b), making their penalty structure more complex than the univariate smooth terms considered here. This additional complexity could further affect calibration beyond what was observed since the approximate significance tests directly depend on the penalties and smoothing parameters in the construction of the covariance matrix and the reference F -distribution.

A related limitation is that the simulations used the thin plate regression splines, which are the default smooths in the ‘mgcv’ framework. Other

spline bases, such as P -splines, outlined by Eilers and Marx (1996), or cubic spline regression, as explored in Section 2.2 of this project, have different penalty structures and null spaces. This is particularly relevant given the observed difficulty in the scenario detecting departures from linearity, where as described previously, the null space of the penalty could directly impact the p -value calibration.

Finally, the study focused mainly on the approximate significance tests implemented in ‘mgcv’. A residual permutation test was explored as a comparison in the most inflated scenario, but was only implemented in the Gaussian case due to computational run time and additional complexity in the permutation process for other response distributions. This limits the extent to which conclusions about the improvement of permutation-based inference can be generalised to the binomial, Poisson and Gamma cases.

Directions for further work

The limitations above suggest several natural directions for further work. The most immediate direction would be to extend the simulation framework beyond the relatively simple independent GAM settings considered in this study. One direction would be to consider generalised additive mixed models, as described by Wood (2006a, p. 316), where the linear predictor is extended to include random effects. This can be written as

$$\mu = \mathbf{X}\beta + \mathbf{Z}\mathbf{b},$$

in correspondence with the matrix form derived in Section 2.4. Here, $\mathbf{Z}$ is a matrix of random effects and $\mathbf{b}$ is a vector of random effect coefficients. This additional term allows for correlation between covariates to be incorporated into the model. Since the additional random effects would introduce further complexity into the penalised likelihood framework, they may also affect the covariate estimates and the effective degrees of freedom used in smooth term inference. Extending the simulations to include correlated error structures would therefore provide useful further insight into how robust the current conclusions are beyond independent GAM settings.

A second direction would be to consider higher dimensional smooth terms. Tensor product smooths would be a natural extension, since Wood (2006b) constructs these with separate penalties and smoothing parameters

for each covariate. This increased complexity of the penalty structure may lead to different calibration behaviour from the univariate smooth terms considered here since it is likely to directly impact the approximations described. This direction is motivated by the findings from Bliss et al. (2011), who found inflated p -values for approximate χ^2 -tests applied to bivariate LOESS smooths in spatial epidemiology, with permutation tests providing more reliable inference. Although their setting differs from penalised tensor product smooths in ‘mgcv’, their findings suggest that higher dimensional smooth terms may introduce additional inferential difficulty. A useful extension would therefore be to directly compare approximate tests for univariate smooths and tensor product smooths directly within the same ‘mgcv’ framework to see if the added complication causes further distortion in p -value calibration.

Future work could also investigate the sensitivity of smooth term inference to the choice of spline basis. There are a range of splines that can be chosen in the ‘mgcv’ framework, so including and comparing alternative spline bases would provide further insight to how sensitive inferential behaviour is to this choice. This would be a particularly relevant addition to the scenario separating linear and nonlinear structure due to the null spaces for each spline likely having a direct impact on inference in this scenario.

Finally, the difficult linear-smooth setting should be studied using alternative inferential procedures. This is motivated by the work of Crainiceanu et al. (2005) and Zhang and Lin (2003), who test specifically to compare parametric against nonparametric terms within a model. Rather than using the approximate F -tests considered here, both papers formulate the corresponding test statistic using a variance-component score test without a standard reference distribution. Zhang and Lin approximate the null distribution using a scaled χ^2 -distribution, while Crainiceanu et al. construct likelihood and restricted likelihood ratio test statistics. This directly addresses one of the key sources of approximation seen in ‘mgcv’ significance test construction where the smoothing parameters are treated as known, whereas these methods acknowledge the variance of smoothing parameters in inference. Zhang and Lin’s simulations show that significance tests for Gaussian cases perform well, but weaker performance in binomial cases, whilst Crainiceanu et al. find that the test statistics have strong power under restricted likelihood ratio tests.

The residual permutation comparison in the Gaussian case also showed that using an empirical null distribution can substantially improve type I error calibration in this setting. However, this improvement comes at a substantial computational cost, since there are hundreds of GAM fittings for each simulated dataset. This makes residual permutation difficult to use routinely in larger simulation studies or applied settings with large datasets. Therefore, a direct comparison between the approximate ‘mgcv’ tests, residual permutation methods, and the specialised tests proposed by Zhang and Lin and Crainiceanu et al. for a wider range of response distributions would be a natural direction for further study. Such a comparison would help determine which methods provide improved calibration without the same computational burden of the permutation test and to what extent they provide better inference for different responses beyond the Gaussian setting.

7 Conclusion

This project has evaluated the reliability of approximate significance tests for smooth terms in generalised additive models reported by ‘mgcv’. The approximate tests were shown to be based on penalised likelihood estimation, large sample assumptions, and the treatment of smoothing parameters as fixed. All of these factors could lead to sources of miscalibration, motivating the simulation-based empirical experiments.

The results suggest that the approximate smooth term tests are generally reliable in simple GAM settings. In type I error simulations, null smooths showed reasonably good calibration for moderate sample sizes and when REML was used as the smoothing parameter estimation method. Some moderate inflation remained at the 10% nominal level, suggesting that larger significance thresholds should be interpreted with caution. The binomial case was least predictable in its type I error calibration across scenarios, showing both conservative and liberal behaviour, which may reflect the greater sensitivity of the large-sample normal approximation for non-Gaussian responses. The power simulations also behaved as expected in simple settings, with increasing sample size and signal strength leading to a greater concentration of p -values near zero. These findings support Wood’s conclusion that the ‘mgcv’ tests can provide useful zero effect tests in relatively simple settings, while extending the comparison across response families, smoothing parameter estimation methods and model complexity. They also address the concerns raised by Young et al. in a penalised GAM framework rather than a LOESS-based setting, and consider both type I error and power.

The main deterioration occurred when the model was required to distinguish between a true linear effect and an additional smooth component of the same covariate. In this case, p -values were noticeably more inflated when compared to simpler null GAM scenarios. Replacing the approximate ‘mgcv’ reference distribution with an empirical null distribution reduced this observed inflation substantially in the Gaussian setting. This suggests the difficulty is strongly connected to the approximate reference distribution used by ‘mgcv’ in this setting, although further work is needed to assess whether similar improvement is possible for non-Gaussian responses.

Overall, the results support the use of ‘mgcv’ smooth term significance

tests as practical approximate inferential tools, especially in simple model structures, moderate sample size and when REML is used in model fitting over GCV. However, they should not be treated as exact evidence of non-linearity. The clearest practical warning from this study is that particular care is needed when using a smooth term test to assess whether a covariate has a nonlinear contribution beyond an already included linear effect. Motivated by the improvement observed in the Gaussian case of the residual permutation comparison, the most important direction for further work is to compare alternative inferential procedures for this specific testing problem for a wider range of responses, particularly methods that may improve calibration without the substantial computational cost of residual permutation.

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A Shared R setup for Monte Carlo error bands and representative ECDF plotting code

This appendix gives the shared R setup used to construct the Monte Carlo standard error bands shown on the type I error ECDF plots. A representative ECDF plotting template is also included, since the same structure was used throughout the type I error and power simulation studies.

Listing 1: Shared R Monte Carlo standard error bands and ECDF plotting template used for type I error simulations.

```
library(mgcv)

# MCSE bands for all plots (given m=5000)
m <- 5000
p_mcse_plot <- seq(0,0.15,length=200)
mcse <- sqrt(p_mcse_plot*(1-p_mcse_plot)/m)
upper <- p_mcse_plot + 1.96*mcse
lower <- p_mcse_plot - 1.96*mcse

# representative ECDF plot
# p_values denotes the vector of simulated p-values from one fitted
# model
plot(ecdf(p_values),
     col = "black",
     xlim = c(0, 0.1),
     ylim = c(0, 0.1),
     xlab = "p-value",
     ylab = "Empirical_CDF")

# reference line of the CDF of U(0,1)
abline(0, 1, lty = 2)

# MCSE bands
lines(p_mcse_plot,upper,lty=3,col="grey40")
lines(p_mcse_plot,lower,lty=3,col="grey40")
```

B Representative code for Type I error Simulation Scenario 1

Listing 2 gives a representative example of the repeated simulation structure used throughout the type I error study. The Gaussian single null smooth term case is shown for varying sample size. The remaining response families follow the same structure with the corresponding response generation and model family.

Listing 2: R code for the Gaussian case of the type I error Simulation Scenario 1 for varying sample size.

```
# set seed for reproducibility
set.seed(101)

# simulation parameters
m <- 5000
lambda <- "GCV.Cp"

# varying sample sizes used
n_values <- c(25,50,100,250,500,1000)

# store results for each sample size
results_list <- vector("list", length(n_values))

# loop over values of n
for (i in 1:length(n_values))
{
  n <- n_values[i]
  p_values <- numeric(m)
  edf <- numeric(m)

  # DGP, generate m datasets
  x_mat <- matrix(runif(n*m, min=0, max=1), nrow=n, ncol=m)
  y_mat <- matrix(rnorm(n*m, mean=0, sd=0.5), nrow=n, ncol=m)

  # fit a GAM to each dataset
  for (j in 1:m)
  {
    sim <- data.frame(x = x_mat[, j], y = y_mat[, j])
```

```

sim.gam <- mgcv::gam(y ~ s(x, k=10),
                    family = gaussian(link = "identity"),
                    method = lambda,
                    data=sim)

s_tab <- summary(sim.gam)$s.table
p_values[j] <- s_tab["s(x)", "p-value"]
edf[j] <- s_tab["s(x)", "edf"]
}

# empirical rejection rates and EDF summary
row_df <- data.frame(
  Response = "Gaussian",
  n = n,
  F1 = mean(p_values <= 0.01),
  F5 = mean(p_values <= 0.05),
  F10 = mean(p_values <= 0.10),
  edf_mean = mean(edf),
  edf_sd = sd(edf)
)

results_list[[i]] <- row_df
}

```

C Representative code for Type I error Simulation Scenario 4

Listing 3 gives the Gaussian version of Simulation Scenario 4, which is the main setting where inflated type I error was observed. This listing shows the construction of the correlated covariate z , the linear response structure, and the fitted GAM containing both the linear z term and the modified smooth $s(z, m = c(2, 0))$.

Listing 3: R code for the Gaussian case of the type I error Simulation Scenario 4 for varying correlation parameter.

```

twopi <- 2*pi

# set seed for reproducibility
set.seed(131)

```

```

# simulation parameters
m <- 5000
sigma <- 0.5
n <- 500

# varying correlation parameter
rho_values <- c(0, 0.3, 0.6, 0.9)

# DGP shared across rho values
x_mat <- matrix(runif(n*m, min=0, max=1), nrow=n, ncol=m)
u_mat <- matrix(rnorm(n*m, mean=0, sd=1), nrow=n, ncol=m)
eps_mat <- matrix(rnorm(n*m, mean=0, sd=sigma), nrow=n, ncol=m)
g_mat <- scale(sin(twopi*x_mat))

# store results for each rho value
results_list <- vector("list", length(rho_values))

for (i in 1:length(rho_values))
{
  rho <- rho_values[i]
  p_values_x <- numeric(m)
  edf_x <- numeric(m)
  p_values_z <- numeric(m)
  edf_z <- numeric(m)
  beta <- numeric(m)

  # construct z and y for current rho
  z_mat <- scale(rho * g_mat + sqrt(1 - rho^2) * u_mat)
  y_mat <- z_mat + eps_mat

  # fit the GAM to each simulated dataset
  for (j in 1:m)
  {
    sim <- data.frame(x=x_mat[, j], z=z_mat[, j], y=y_mat[, j])

    sim.gam <- mgcv::gam(y ~ z + s(x, k=10) + s(z, k=10, m=c(2,0)),
                        family = gaussian(link = "identity"),
                        method = "REML",
                        data=sim)
  }
}

```

```

    s_tab <- summary(sim.gam)$s.table
    p_values_x[j] <- s_tab["s(x)", "p-value"]
    edf_x[j] <- s_tab["s(x)", "edf"]
    p_values_z[j] <- s_tab["s(z)", "p-value"]
    edf_z[j] <- s_tab["s(z)", "edf"]
    beta[j] <- coef(sim.gam)["z"]
  }

  # empirical rejection rates and summary statistics
  row_df <- data.frame(
    rho = rho,
    F1x = mean(p_values_x <= 0.01),
    F5x = mean(p_values_x <= 0.05),
    F10x = mean(p_values_x <= 0.10),
    edf_meanx = mean(edf_x),
    edf_sdx = sd(edf_x),
    F1z = mean(p_values_z <= 0.01),
    F5z = mean(p_values_z <= 0.05),
    F10z = mean(p_values_z <= 0.10),
    edf_meanz = mean(edf_z),
    edf_sdz = sd(edf_z),
    beta_mean = mean(beta),
    beta_sd = sd(beta)
  )

  results_list[[i]] <- row_df
}

```

D Full code for Type I error Simulation Scenario 4 residual permutation comparison

Listing 4 gives the residual permutation procedure used for the Gaussian version of Simulation Scenario 4. The fitted null model removes the tested smooth term $s(z, m = c(2, 0))$, while retaining the linear z term and $s(x)$ smooth term. Residuals from this null model are permuted to generate pseudo-responses, and the full model is then refitted to obtain an empirical null distribution for the F -statistic of $s(z, m = c(2, 0))$.

Listing 4: R code for the Gaussian type I error Simulation Scenario 4 residual permutation test comparison.

```
# set seed for reproducibility
set.seed(401)

# simulation parameters
m <- 5000
n <- 500
rho <- 0.6
sigma <- 0.5
no_perm <- 499
beta_true <- 1
twopi <- 2*pi

# DGP
x_mat <- matrix(runif(n*m, min=0, max=1), nrow=n, ncol=m)
u_mat <- matrix(rnorm(n*m, mean=0, sd=1), nrow=n, ncol=m)
g_mat <- scale(sin(twopi*x_mat))
z_mat <- scale(rho * g_mat + sqrt(1 - rho^2) * u_mat)
eps_mat <- matrix(rnorm(n*m, mean=0, sd=sigma), nrow=n, ncol=m)
y_mat <- beta_true * z_mat + eps_mat

# store mgcv and residual permutation p-values
p_values_z <- numeric(m)
p_perm_z <- numeric(m)

# loop over number of simulation replications
for (j in 1:m)
{
  if (j %% 50 == 0) print(j)

  sim <- data.frame(x=x_mat[, j], z=z_mat[, j], y=y_mat[, j])

  # full model: observed F-statistic for s(z)
  sim.gam <- mgcv::gam(y ~ z + s(x, k=10) + s(z, k=10, m=c(2,0)),
    family = gaussian(link = "identity"),
    method = "REML",
    data = sim)

  s_tab <- summary(sim.gam)$s.table
```

```

p_values_z[j] <- s_tab["s(z)", "p-value"]
F_obs <- s_tab["s(z)", "F"]

# null model: removes s(z), retains linear effect of z and s(x)
sim.gam_null <- mgcv::gam(y ~ z + s(x, k=10),
                        family = gaussian(link = "identity"),
                        method = "REML",
                        data = sim)

y_fitted_null <- fitted(sim.gam_null)
residual_null <- residuals(sim.gam_null, type = "response")

F_perm <- numeric(no_perm)

# residual permutation under the fitted null model
for (b in 1:no_perm)
{
  sim_perm <- sim

  # generate pseudo-response under the null hypothesis
  sim_perm$y_perm <- y_fitted_null + sample(residual_null,
      replace=FALSE)

  # refit full model to pseudo-response
  perm.gam <- mgcv::gam(y_perm ~ z + s(x, k = 10) + s(z, k = 10,
      m = c(2, 0)),
                      family = gaussian(link = "identity"),
                      method = "REML",
                      data = sim_perm)

  s_tab <- summary(perm.gam)$s.table
  F_perm[b] <- s_tab["s(z)", "F"]
}

# permutation p-value
p_perm_z[j] <- (1 + sum(F_perm >= F_obs)) / (no_perm + 1)
}

row_df <- data.frame(
Model = c("GAM", "PERM"),

```

```
F1z = c(mean(p_values_z <= 0.01), mean(p_perm_z <= 0.01)),  
F5z = c(mean(p_values_z <= 0.05), mean(p_perm_z <= 0.05)),  
F10z = c(mean(p_values_z <= 0.10), mean(p_perm_z <= 0.10))  
)
```

E Full simulation code

All simulations were run using R version 4.3.1 and ‘mgcv’ version 1.8-42. The full R code used to generate all simulation results and figures is available at: <https://github.com/mattcrean/GAM-Simulation-R-Code>